UNIT 4

Natural Resources: Problems and Prospects

The, Earth provides enough to satisfy every body's needs but not for any body's greed.

Mahatma Gandhi

Objectives

- Introduction
- Classification or Types of Resources
- Forest Resources
- Water Resources
- Mineral Resources
- Food Resources
- Energy Resources
- Land Resources

4.1 INTRODUCTION

Everything available in our environment which can be used to satisfy our needs, provided it is technologically accessible, economically feasible and culturally acceptable can be termed as 'Resource'. Thus any part of our environment such as land, air, water, wild life, minerals, forest and even human population that human beings themselves can utilise to promote their welfare by their intellectual efforts and technologies may be regarded as resource. In other words one can say that the five basic ecological variables i.e., energy, matter, space, time and diversity together can be known as resource. These resources form the backbone of the economy and prosperity of a nation.

4.2 CLASSIFICATION OR TYPES OF RESOURCES

Natural resources can be classified into two categories on the basis of origin:

(i) *Biotic Resources*

These are obtained from the biosphere and have originated from some living organisms or have life such as human beings, flora and fauna, fisheries, livestock and some fossil fuels like coal, petroleum etc. The biotic resources could be renewable or non-renewable type.

(ii) Abiotic Resources

All those things which are of non-living origin are termed as abiotic resources for example minerals, rocks, metals, soil, water etc.

Resources can also be classified on the basis of duration and availability of resources into two categories:

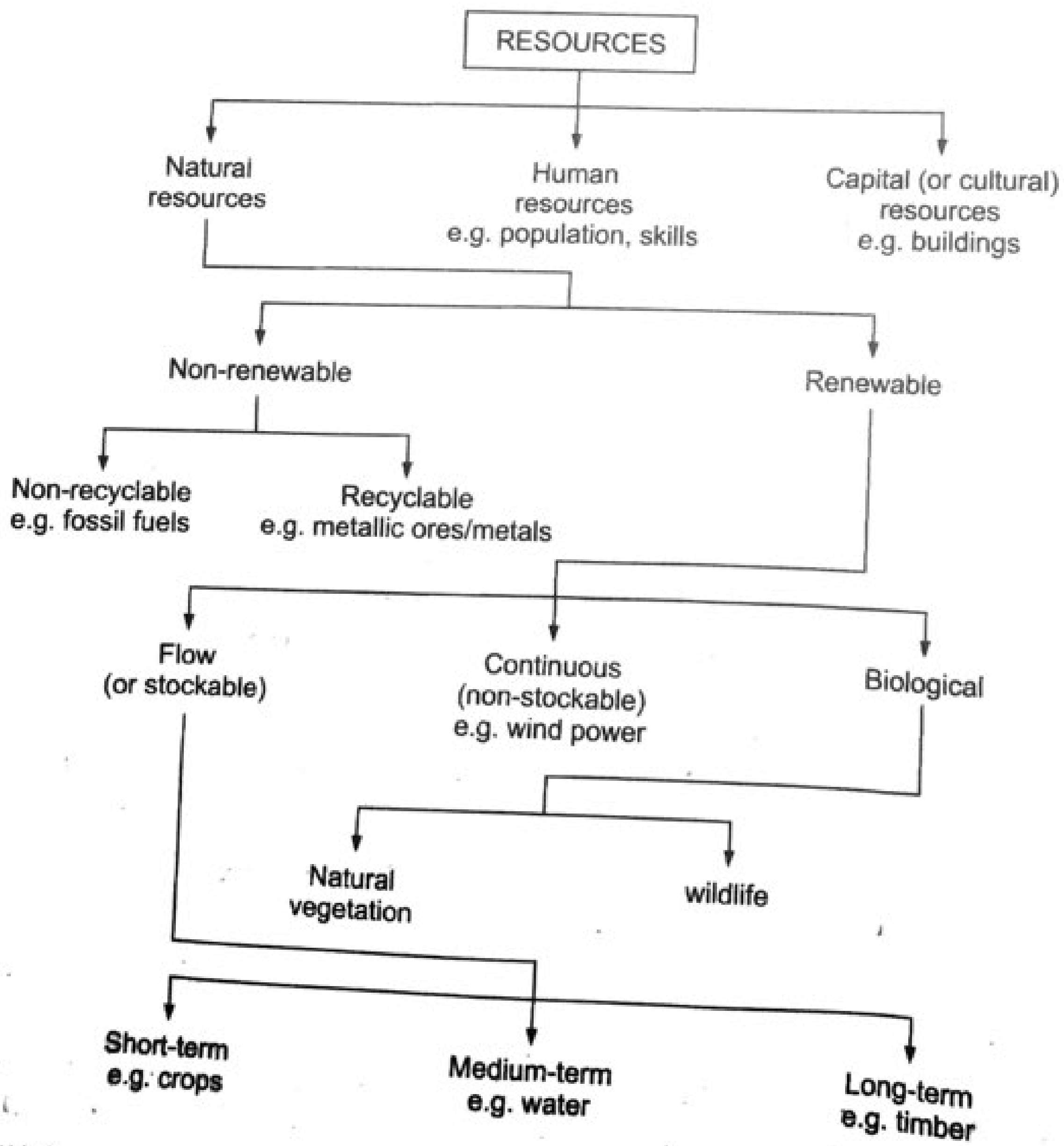
(i) Renewable Resources

The resources that are non-exhaustible in nature and can be renewed or reproduced by physical, chemical or mechanical processes are known as renewable or replenishable resources. For example, solar and wind energy, water, forests and wildlife etc.

The renewable resources may further be divided into flow, continuous and biological.

(ii) Non-renewable Resources

These occur over a very long geological time. Mineral and fossil fuels are examples of such resources. These resources take millions of years in their formation. Fossil fuels by no means can be recycled and get exhausted with their use, while some of the resources like metals are recyclable and reusable.



Resources are vital for any developmental activity. But irrational consumption in the relentless march towards development have plundered the earths natural resources which have finally threatened human survival on earth. Thus conservation i.e. judicious and planned use of resources becomes essential as it aims at sustainable benefit to the present generation. It also maintains a potential to meet the needs and aspirations of future generations.

4.3 FOREST RESOURCES

The forest is a complex ecosystem consisting mainly of trees that shield the earth and support innumerable life forms. They play a major role in enhancing the quality of environment. They clean the air, cool it on hot days, conserve heat at night, and act as excellent sound absorbers. Forests also control soil erosion, regulate stream flow, support a variety of industries, provide livelihood for many communities and offer opportunities for recreation. They also are a natural habitat for the wildlife. The National Forest Policy (1988) emphasises the role of forests in maintaining the life support system.

The forests of India are ancient in nature and composition. They are rich in variety and shelter a wide range of flora fauna. India possesses a distinct identity, not only because of its geography, history and culture but also because of the great diversity of its natural ecosystems. The panorama of Indian forests ranges from evergreen tropical rain forests in the Andaman and Nicobar islands, the Western Ghats and the north-eastern states, to dry alpine scrub high in the Himalaya to the north. Between the two extremes, the country has semi-evergreen rain forests, deciduous monsoon forests, thorn forests, subtropical pine forests in the lower montane zone and temperate montane forests.

For the purpose of administration, forests in India are classified into three types, namely:

- Reserved forests
- Protected forests
- Unclassed forests

Reserved forests are those which are permanently earmarked either to production of timber or other forest produce and in which right of grazing and cultivation is seldom allowed. In **protected forests** these rights are allowed subject to a few minor restrictions. **Unclassed forests** consist largely of inaccessible forests.

Forests are also classified according to density namely: **dense**, **open** and **mangrove**.

Dense forests are those which have the crown cover density greater than 40%; while forests having crown cover densities between 10% to 40% are included in the **open forests**. The vegetated areas, where the crown density is less than 10% are known as scrub lands and are usually not included in the forest land.

Mangrove forests are mainly found along stretches of tropical ocean. They usually grow in places near quiet ocean waters. These forests often grow in shallow waters along bays, lagoons and river mouths. The thousands of stilt-like roots of a mangrove tree catch silt and sand, which piles in the shore water. These roots slow down the current and help settle the silt. The mangroves, thus, help in building up dry land.

In India, at present, forest areas cover about 2.9 billon hectares of land which is about 22 per cent of the total geographical area. Of the total forest cover, 54.5 per cent are reserved, 29.2 per cent protected and 16.4 per cent unclassed forests (Fig. 4.1). About 59 per cent of the forest area are dense, 40 per cent open, and less than one per cent is mangrove forests (Fig. 4.2).

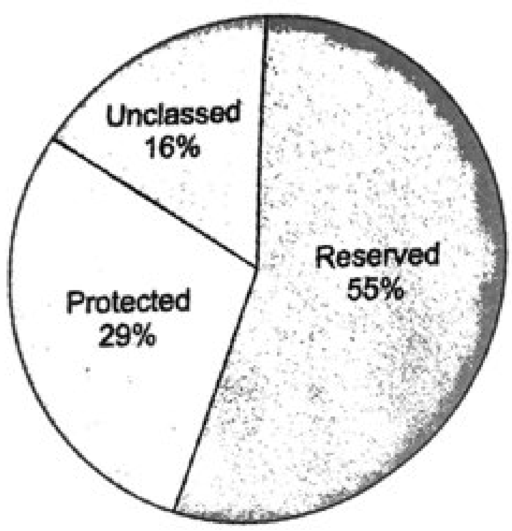


Fig. 4.1 Types of forests in India

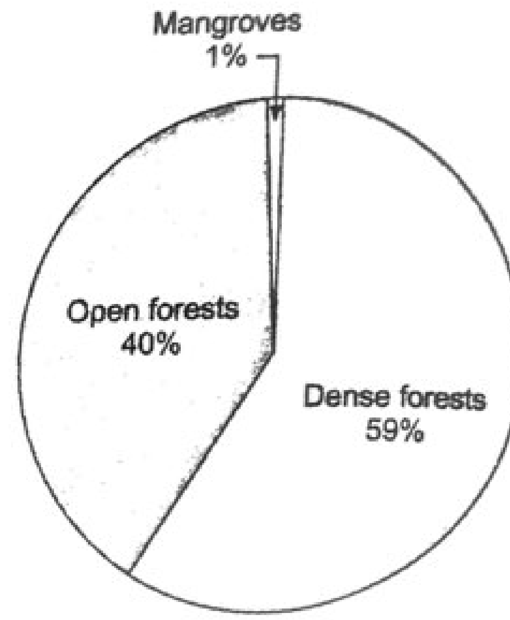


Fig. 4.2 Forest cover by density class.

4.3.1 Importance of Forests

Functions of a forest may broadly be classified into following three categories:

(*i*) **Protective functions:** These include the protective role of forests against soil erosion, droughts, floods, intense radiation etc.

(*ii*) **Productive functions:** Forests are the source of wood and many other products like gums, resins, fibers, medicines, honey, pulp, paper etc.

(*iii*) **Accessory functions:** These include the role of forests in recreation, aesthetics and as adobe of many animals and tribal people. It maintains the biodiversity of nature.

Thus forests are important both ecologically and economically.

Ecological Significance

The forests help in balancing oxygen and carbon dioxide level in atmosphere, regulate earth's temperature regime and hydrological cycle. It increases local precipitation and water holding capacity of soil, thus preventing drought situation. Vegetation cover provided by forest impedes the velocity of run off on soil surface, checks soil erosion, silting and landslides thus reducing the danger of flood. The litter derived from fallen leaves maintains fertility of soil by returning the nutrients in form of humus. Forests also act as refuge for wild animals and provide protection to them against strong, cold or hot and dry winds, solar radiation, rain and enemies.

Economic Significance

Wood which has various applications in domestic and industrial processes, is the chief product of forest. Wood when used as fuel, has certain advantages over coal as its sulphur and ash contents are very low but, at the same time, excessive use of fuelwood means pressure on forests which have many functions to serve. Wood may also be converted to solid and gaseous fuels. Timber is an important material in building construction and day-to-day uses. Forests provide raw material for various wood based industries viz., pulp and paper, composite wood, rayon and other man-made fibers, sport goods, furniture, matches etc. Miscellaneous products like bamboos, resins, gums, some oils, fibers medicines etc. are also obtained from forests.

4.3.2 Forest Cover of the Earth

It is accepted that, ideally, the Earth should have a forest cover of one third of its land area, which should extend to 66 per cent forest cover in the hilly regions. At the beginning of the twentieth century, the Earth had a forest cover of 5 billion hectares (38.5 per cent of the land area). But presently it is estimated that the forests occupy an area of 2.9 billion hectares, which works out to just 22 per cent of the land area. This figure represents the world average. It is a fact that forest loss into enormous proportions is mainly in developing countries. It is estimated that the developing world is losing 6.5 per cent of its forests per decade. This is in contrast to the industrialized world, which is gaining forest land.

Forest Cover in India

India's forests cover an area of approximately 67.5 million hectares, 20.6 per cent of the geographical area of the country. These forests are classified as

Table 4.1

	Area km^2	*Percent of geographical area of the country*
(i) Dense forests (with a canopy cover greater than 40% of the wooded area)	416,809	12.7
(ii) Open forests (with a canopy cover of 10-40% of the wooded area)	258,729	7.9
Total		20.6
(iii) Scrubs	47,318	1.4

Source: Dept. of Forests, Govt. of India 2003-2004 data.

Forest Cover per Person

The forest cover of India is comparable with that of the world average. But the forest cover per person is meagre as the country has about 17 per cent of the world's population of 6.2 billion. The forest cover of India works out to 634 m^2/person, which, compared to that of the United States, Canada, Russia and the Latin American countries, is very small.

There has been a fast disappearance of tropical forests in our country, at a rate of about 7.3 million ha per year. The major cause for the depletion of forest cover is the expansion of agricultural practices in the country. The areas with rich biological diversity were shifted by the modern practices into the areas of cultivation with a few species of hybrid crop plants.

Another cause for over exploitation of forest resources due to reclamation activities of building dams, factories, highways, mining operations etc. Over population in third world countries is often cited as major cause of over exploitation of forest resources.

4.3.3 Deforestation

Forests are the biggest source of materials and land for modern society. Deforestation has become inevitable for supporting human activities. The need of fuelwood and timber for construction and furniture continues to increase as population is expanding.

India has 67.5 million hectares of forest land out of a total geographical area of 329 million hectares. It is on record that the country had 80 per cent forest cover 5000 years ago which has now been reduced barely to 22 per cent as discussed above. Per capita forest land is 0.1 hectare compared to the world average of 1 hectare. This shows the poorest record of India. We are losing 1.3 million hectares of forest cover every year and at this rate of deforestation we are likely to lose all the forest cover and face total disaster within the next 25 years.

During 1951-2000 i.e. almost 50 years after Independence, India witnessed maximum loss of forests, namely, 6.0 million hectares of forest area. This overconsumption has been due to population explosion (both human and livestock population) accompanied by rapid urbanization and industrialization as well as change of life style. There has been tremendous pressure on forest systems to meet ever-increasing demands on fuel or fire wood, timber and pulp wood for housing, furniture, agricultural implements, railway slippers, sports goods, ship and boat building, paper and pulp industries etc. Annual consumption of fire wood in the country is about 170 million tons while 10-15 hectares of forest cover is destroyed each year to meet fuel requirements. The consumption of fire wood is 80 per cent in villages and 20 per cent in urban areas. There is a shortfall of about 195 million m^3 in the demand and supply of fuel wood. Similarly the gap for timber between demand and supply is about 15 million m^3.

The forests are subjected to at least five times more pressure than what they can withstand. **The imbalance between demand and production of fuelwood and timber is one single factor that has contributed most to the depletion of forests in the country.**

Excessive grazing, frequent fires, and other biotic factors have adversely affected the natural regeneration of our forests. Nearly 73.6% of the forest area does not have regeneration of important species, and the percentage of forest area with no regeneration varies from 15% (West Bengal) to 90% (Jammu and Kashmir). These are indicators of deteriorating situation leading to progressive degradation of forests. The forest area lost due to various reasons during the period 1952-80 is inferred to be nearly 43.280 km^2 with an average loss of 1550 km^2 annually. The impact of agriculture on the denudation of forests has been the maximum as exemplified by the diversion of the largest forest area (26.230 km^2) to agriculture during this period. However, on the enactment of the Forest Conservation Act in 1980, the forest area diverted for non-forest uses between end-1980 and mid-1993 was 3040 km^2, reducing the rate of diversion to approximately 250 km^2 per annum. Nearly 7000 km^2 of forest land was encroached upon between 1951 and 1982.

Shifting Cultivation (Jhumming): A low-energy budget and a low-investment system of subsistence husbandry — is practised in about 16 states of India, and is most prevalent in the seven north-eastern states and Orissa. Since there is not enough area for the system to function, forest degradation is rampant.

Forest fire: It is one of the most dangerous sources of damage to forests and wildlife. On an average, 53.1% of the forest area is affected by fire and it ranges from as low as 6.8% to as high as 97%. About 8.92% of the forest area is being affected by frequent fires and 44.25% by occasional fires.

The pressures of **increasing livestock population** and migratory graziers are a potent force, responsible for the continual degradation of forests, and grasslands, causing widespread devastation. It is estimated that currently over 100 million heads of livestock or 110 million cow units graze in the forests of the country. Considering the degraded state of the forests, and a minimum requirement of 2 ha of land for one cow unit, the grazing pressure on the forests is more than three times, which is far beyond the carrying capacity of this resource.

4.3.4 Consequences of Deforestation

The process of deforestation results in many undesirable environmental impacts at multiple scales. The local impacts include ***decreasing soil stability, increasing erosion*** *and* ***sediment transport*** into streams, ***reduction in biodiversity*** through loss of habitat and ***alterations to microclimates*** that typically increase local temperatures because of loss of vegetation. ***Degradation of air quality*** is often at the regional scale if deforestation is being driven by burning down slash as in shifting cultivation. This promotes high levels of ***air pollution*** (mainly particulate matter, carbon monoxide, sulphur oxides and nitrogen oxides) that is harmful to both human and wildlife. Deforestation can also produce impacts on global scale. Cutting and burning of large forest tracts liberates large amounts of carbon and ***increase levels of green house gas,*** carbon dioxide

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4.3.5 Conservation of Forests

Over the past few decades both the government and local communities have resisted the temptation of indiscriminate exploitation of forest resources, and thus conservation of forests in India has become significant. Of further significance is the fact that forest conservation in India started well before the global concerns of climate warming and decline of biodiversity, and thus represents an indigenous response to environmental concerns and not a response to external pressure.

The factors that have contributed to forest conservation in India are as follows:

The forest Conservation Act, 1980. This bans off unnecessary and avoidable forest conversion to non-forest use. Only 46000 ha of forest have officially (FSI 1988) been diverted to non-forest use in the 7 years after 1981, mainly for river valley projects, compared to 4.3 Mha officially diverted between 1951 and 1980.

Compensatory Afforestation Any permission given for forest conversion must be accompanied by compensatory afforestation on degraded land.

Establishment of National Parks and Wildlife Sanctuaries 12.6 Mha of forest has been dedicated to nature reserves.

Afforestation Programme India has launched a large afforestation programme in degraded forests, village commons, and farm land. The area afforested in India from 1980 to 1990 was 14.4 Mha.

Joint Forest Management: Under the legislation of 1988 and 1990, a joint programme involving local communities and the Forest Department has been launched with the aim of protecting and promoting natural regeneration in degraded forest lands. Under such a programme, in three disricts of West Bengal, 191750 ha of degraded forest land was revegetated between 1984 and 1990.

Removal or Reduction of Subsidies to Industry Earlier, industry enjoyed a guaranteed supply of timber at low or nominal rates from forests. With the reduction in subsidies, industry is expected to use forest resources efficiently.

Environmental Movements, Free Press and Judiciary Environmental movements, starting with the 'Chipko movement' in the Himalayas, have spread to several states. India's free press and judiciary is generally progressive and supports forest conservation efforts and acts as a check on any indiscriminate deforestation.

National Forest Policy, 1988

The following are the main objectives outlined in the National Forest Policy of India, 1988:

- Maintenance of environmental stability through preservation and where necessary, restoration of the ecological balance that has been adversely disturbed by serious depletion of the forests of the country.
- Conservation of the natural heritage of the country by preserving the remaining natural forests along with their vast variety of flora and fauna, which represent the remarkable biological diversity and genetic resources of the country.
- Control of soil erosion and denudation in the catchment areas of rivers, lakes, and reservoirs in the interest of soil and water conservation, for mitigating floods and droughts and for the retardation of siltation of reservoirs.
- Checking the extension of sand dunes in the desert areas of the Rajasthan and along the coastal tracts.
- Increasing substantially the forest/tree cover in the country through massive afforestation and social forestry programmes, especially on all denuded, degraded, and unproductive lands.
- Meeting the requirements of fuel wood, fodder, minor forest produce, and small timber of the rural and tribal populations.
- Increasing the productivity of forests to meet essential national needs.
- Encouraging efficient utilization of forest produce and maximizing substitution of wood.
- Creating a massive people's movement, with the active involvement of women, for achieving these objectives and to minimize the pressure on existing forest areas.

4.3.6 Timber Extraction

Wood is an important component of a forest thus its removal has immediate implications on it. It is commonly used as a building material, for making furniture as a fuel. Good quality and durable wood which is used as a building material and for making furniture is commonly known as *Timber*. With growth of population and industrialization, the demand for timber is increasing, which has led to the loss of natural forests.

Impact of Timber extraction

Loss of biodiversity: Removal of forest cover leads directly to a loss of animal habitat and plant species thus a decline in biodiversity. The increased spring run-off coupled with summer drought, soil erosion and land-slips results in hotter summers and cooler winters. The drying of the forest areas lead to an increased risk of forest fires.

Loss of Carbon Storage Capacity: Carbon dioxide emissions in the atmosphere due to intense use of fossil fuels during the course of industrialization has led to global warming. The present concentration of CO_2 in the atmosphere is 387 ppm. Forests, apart from being a source of

CO_2, also provides opportunity for reducing atmospheric CO_2 concentration. Mitigation potential of forestry sector at global level is estimated to be 60 to 87 Gt C during the period 1995 to 2050 (Brown, 1996). Carbon can be removed through the growth of forests. (CO2 sequestration discussed in detail in the next unit) Trees act as a sink and thus, forests represent the largest carbon storage mechanisms in the global carbon cycle. When the forests are destroyed the carbon storage capacity is lost. Moreover the additional carbon is released into the atmosphere through decay and burning.

4.3.7 Case Study

CASE STUDY

Chipko-Movement

Chipko-movement started in Nineteen Seventees in a small hill village of the upper reaches of Himalayas. Tribal people of Tehri-Garhwal district of Uttarakhand realised the importance of forests and decided against giving its products to the people of other areas. They stood against the ruthless butchery of nature and the axes of greedy contractors. At the initial stage of the movement (in December, 1972), the women of Advani village in Tehri-Garhwal protested against indiscriminate felling of trees. In March 1973, a sports goods factory was to cut the Ash trees near the village Mandal in Chamoli district. The local people prevented the same by hugging (vern. Hindi Chipko) the marked trees. In 1994, a group of women led by Gaura Devi successfully prevented felling of trees near village Reni. The movement became famous in 1978 when the women of Advani village in Tehri-Garhwal faced police firing and later courted arrest. The Chipko Movement spread slowly to all nearby areas under the leadership of Shri Sunderlal Bahuguna of silyara in Tehri region and Shri Chandi Prasad Bhatt of Gopeshwar.

Chipko movement has challenged the old belief that forests mean only timber. It emphasises that important gift of tree to us is not timber, but soil, water and oxygen. The slogan of Chipko movement is infact planting five Fs-food, fodder, fuel, fiber and fertiliser trees.

The tribals of the other parts of the country got inspiration from Chipko movement and started similar movements in their areas. One such movement reached to Karnataka as Appiko movement under the leadership of Pandurange Hegde. The other parts of the country where this movement reached are – Santhal Parganas, Chhatisgarh, Thane and the Aravali regions.

Source: Brown S., 1996, The role of forest ecosystems and forest management in the global carbon cycle, Springer-Verlag, New York.

4.3.8 Dams and Their Effect on Forest and Tribal People

In India, national development has been largely equated with economic growth and surplus. Large, centralized industries, irrigation project. have been symbols of such development, which through the process of industrialization promised to set the country on the path of modernization and development. One of the inevitable outcomes of this has been massive environmental degradation and 'development induced displacement'.

Immediately after independence, a series of large dams were planned and built on some of the major rivers in India. Large dams promised to solve the problem of hunger and starvation by providing irrigation and boosting food production, controlling floods and providing much needed electricity for industrial development. It was this grand promise that prompted Pandit Nehru, our first Prime Minister, to call dams "*secular temples of modern India*". Environmental and social costs of such large dams were thought to be an inevitable price that one had to pay for such development. *Socio ecological costs of large dams were grossly underestimated and largely ignored.*

A large dam inevitably alters the course of nature and because of the play of complex and largely unknown forces, *the ecological consequences are very serious*. The forest cover bears the brunt of dam construction in a severe manner. The forest area which is supposed to get submerged, is cleared off by the contractors. The people who are displaced also are rehabilitated on the forest land that too is acquired and cleared off. The forest is also cleared for approach roads, offices, residential quarters and for storage of construction material. With the reduction in the forest cover and the entry of people for dam construction, the pressure on remaining forest increases. The people's need for firewood leads to further denudation. The construction of a dam, therefore has a ***multiplier effect***. The large number of *tribal people get displaced* and affected by dam construction. The estimates show that dams are the single largest cause of displacement accounting for 75 to 80% of the total displacement.

Table 4.2. Independent estimate of total person displaced by dams during 1950-90 (in lakhs)*

Category	Number (in lakhs)	Percentage (%)**
Total number of persons displaced	164	100%
Total number resettled	41	25%
Backlog	123	75%
Total number of tribal persons displaced	63.21	38%
Total tribal persons resettled	15.81	25%
Backlog	47.40	75%

Source: Amita Patwardhan, Dams and tribal people in India, **Dams and** Development: A new framework for decision-making, 2000.

** Percentage is out of total tribal persons displaced.

It has been estimated that large majority of the people are Scheduled Tribes and Scheduled Castes. The tribal communities are often forced to migrate to urban slums in search of employment or become landless labourers. They never share the benefits of the projects which have displaced them, be it irrigation or electricity.

Anti dam movement in India also has a long history. The first struggle against a dam was fought in early 1920s by peasants in Western Maharashtra who opposed the Mulshi dam built by the Tatas. In recent past, from 1970s till today several anti dam struggles have posed a major challenge to dam building in India. Silent valley was one of the first success stories where a dam threatening to submerge rich forest was scrapped. One of the longest and most well known struggle against large dams was in the Narmada valley against the Sardar Sarovar Project, known as Narmada Bachao Andolan (NBA), led by Medha Patkar. The struggle began in the mid-1980s and it had a profound impact on the debate on large dams and role of people's movement in India. NBA, along with other movements, has brought some of the most of the most fundamental issues regarding equality, social justice and the notion of development of the forefront.

4.3.9 Case Study

CASE STUDY

Narmada Valley Project

Narmada is the largest western flowing river arising from the Kantaka plateau in shahdol district. Millions of people live in the basin of this river and use the basin as a resource for their livelihood. Narmada Valley Project was planned with a view to using its water utilization capacity to an optimum level for the benefit of the people. The plan included construction of two large dams (Sardar Sarovar and Narmada Sagar Projects) and many small dams. In would lead to generation of electric power, irrigation of million hectare of land in Gujarat, Madhya Pradesh and Rajastan and making availability of drinking water. The entire project and the World Bank's role in supporting it till 1993 created controversy. While the project brought benefits as mentioned above, it threatened the livelihood of thousands of people in the area to be flooded by Sardar Sarovar Dam. Also, the project would have enormous environmental impacts. The dam has been designed to divert water from Narmada river into canal and irrigation systems.

The anti-dam stir started in the valley in 1985 in Gujarat, Maharashtra and Rajasthan to stall the building of Sardar Sarovar and Narmada Sagar Dams. In 1989, Narmada Bachao Andolan (NBA), which had the support of other NGO's, became a major force to resist the government efforts that would result in the destruction of environment and bring miseries to people especially tribals. NBA has been holding demonstrations, courting arrests led by Medha Patkar, demanding in 1990, that the project be suspended. NBA succeeded in influencing the suspension of Japanese aid. Baba Amte, a noted social worker, supported NBA and started a march from Madhya Pradesh to the site of the dam.

4.4 WATER RESOURCES

It is believed that life first originated in water before it invaded land. Water is in fact a precondition of life and essential for the sustenance of all forms of life, food production, economic development and for general well being.

The ***primary source of water on the earth is precipitation*** that comes in the form of rain and snowfall. India receives annual precipitation of about 4000 km^3 including snowfall.* Out of this, monsoon rainfall is of the order of 3000 km^3. India is gifted with a river system comprising more than twenty major rivers with several tributaries. The rivers are the main source of surface water in India apart from ponds, tanks and reservoirs etc. The mean annual flow of the Indian rivers is estimated to be about 1869 billion cubic meters (bcm). About 690 bcm or 37% of it is available for use. The Indus, the Ganga and the Brahmaputra carry 60% of the total surface water. On the basis of their hydrology, the Indian rivers are divided into two categories: the Himalayan and Peninsular. Most of the Himalayan rivers have their sources in the glaciers and snowfields and therefore are perennial in nature, while the peninsular rivers depend entirely on monsoon rains and hence are seasonal. Thus, the peninsular rivers demand storage of water for irrigation and power generation.

Apart from the water available in the various rivers of the country, ***the ground water is also an important source of water for drinking, irrigation, industrial uses etc.*** It accounts for about 80% of domestic water requirement and more than 45% of the total irrigation of the country. As per the international norms, if per capita water availability is less than 1700 m^3 per year then the country is categorized as water stressed and if it is less than 1000 m^3 per capita per year then the country is classified as water scarce. In India per capita surface water availability in year 2001 was 1900 m^3 and it is projected to reduce to 1400 m^3 by the year 2025 (Kumar et. al., 2005). This water scarcity is an outcome of large and growing population and consequent greater demands for water. An ever increasing growth in population alongwith intensive industrialization and urbanization means more water not only for domestic use but also to produce more food.

Hence to facilitate higher food production, water resources are being over exploited to expand irrigated areas and dry-season agriculture alongwith other developmental activities like production of hydroelectricity, water supply for industrial and domestic use, recreation, inland navigation and fish breeding.

***Source:** Rakesh Kumar et al., Current Science, 89(5), 2005.

4.4.1 Over-utilization of Surface and Groundwater

Worldwide the largest use of water is for irrigation (70%), second is for industry (20%) and third is for direct human use (10%). The total annual requirement of water in India for various sectors has been estimated in the table 4.3.

Table 4.3. Annual water requirement for different uses. (in km^3)

Use	1997-98			2010			2025			2050		
	Surface Water	Ground Water	Total	SW	GW	Tot.	SW	GW	Total	SW	GW	Tot.
Irrigation	318	206	524	339	218	557	366	245	611	463	344	807
Domestic	17	13	30	24	19	43	36	26	62	65	46	111
Industries	21	9	30	26	11	37	47	20	67	57	24	81
Power	7	2	9	15	4	19	26	7	33	56	14	70
Inland Naviation	—	—	0	7	—	7	10	—	10	15	—	15
Environment-Eclogy	—	—	0	5	—	5	10	—	10	20	—	20
Evaporation losess	36	—	36	42	—	42	50	—	50	76	—	76
Total	399	230	629	458	252	710	545	298	843	752	428	1180

Source: Integrated water resources development - A plan for action, Report of the National Commission for Integrated Water Resources Development, Ministry of Water Resources, New Delhi, 1999.

In view of the existing status of the water resources and increasing demands of water for meeting the requirements of the rapidly growing population of the country, the need of the hour is to conserve and manage our water resources. We have to safeguard ourselves from health hazards, to ensure food security, continuation of our livelihoods and productive activities and also to prevent degradation of our natural ecosystems. Over exploitation and mismanagement of water resources will impoverish this resource and cause ecological crisis that may have serious impact on our lives.

4.4.2 Consequences of Over Exploitation of Surface Water Resources

To trap and control flowing surface water, dams and reservoirs are build. The stored river water alongwith rainwater in a dam or a reservoir is used for irrigation, electricity generation, water supply for domestic and industrial uses, flood control, recreation, inland navigation and fish breeding.

These dams and reservoirs have enormous ecological impacts. When a river is dammed, valuable fresh water habitats are lost. Sometimes the river's flow is diverted to cities or croplands, which ***affects the fishes and the aquatic organisms***. The ***wildlife*** that depends on the water or the food chains involving aquatic organisms is also ***adversely affected***. Sometimes fishes swim up river from the ocean to *spawn* are seriously affected by reduced water level. The reservoirs which are created on the ***floodplains also submerge*** the existing vegetation and soil leading to its decomposition over a period of time. Due to diversion in the river's flow, estuaries are also affected because less fresh water enters and flushes the estuary. Consequently salt concentration increases profoundly *affecting the estuary's ecology*.

4.4.3 Consequences due to the Depletion of Groundwater

If the groundwater withdrawals exceed its recharge, the ***water table falls***. When the water table falls, the springs and seeps start drying up and even the stream and the rivers are affected. Thus ***surface water diminishes*** and this creates the same results as the diversion of surface water. Over the ages, groundwater has leached cavities in the ground. The water fills these spaces and helps support the overlying rock and soil. Due to fall in water table this support is lost and there may be a gradual settling of the land, known as ***land subsidence.***

Land subsidence may cause building foundations, roadways, and water and sewer lines to crack. In coastal areas this phenomenon may cause flooding.

Another problem resulting from dropping water tables is ***saltwater intrusion.*** In the coastal areas, the springs of overflowing groundwater may lie under the ocean. As long as a high water table is maintained a sufficient head of pressure is available in aquifer and freshwater will flow into the ocean. Thus wells in the coastal areas have fresh water. Lowering of water table at a rapid rate reduces pressure in the aquifer and thus permitting salt water to flow into the aquifer and hence into wells.

4.4.4 Water Conservation and Management

Water resources can serve more people by wise use and good management. Every human being has the responsibility to conserve water used in the various activities. Several approaches to conserve and manage the water resources are:

- ***Avoid Polluting***—Pollution makes water unfit for use. Renewing takes time and nature may not be able to renew it if the pollution is bad. Properly dispose off oil so it does not get into water. Use pesticides sparingly and donot use excess of detergents and soap.
- ***Dispose off properly***—Proper disposal of wastewater helps protect natural supplies. Wastewater can be partially renewed in treatment plant facilities.
- ***Install conservation practices***—Many approaches can be used to conserve water supplies and quality. Approaches that conserve soil also conserve water. Terraces, ponds, and mulches can be used to reduce water runoff. Factories can seek more efficient ways of using water.
- ***Have good equipment***—Pipes, pumps, and other facilities should be free of leaks. Leaky equipment wastes water. It also costs more to operate a leaky system because more water must be pumped just to have enough to do what is needed. More energy is needed to power the pumps.
- ***Reuse***—Water used for one purpose can often be used for other purposes before it is released. The additional uses may help clean

the water. An example is using wastewater to raise fish edible plants. Both activities remove nutrients in wa food manufacturing operations.

- ***Renew used water***—Renewing wastewater is helping job. It may involve filtering to remove solid materia reservoirs, it might include promoting the growth o processes occur, such as the nitrogen cycle.
- ***Efficient use of water***—Everyone can make better u consuming a little less water for the daily chores.

Water Harvesting for Conservation

In India, from the ancient times there exists an extraordin water-harvesting system. People have in-depth knowle regimes and soil types and thus have developed various harvest rain water, ground water, river water and flood w with the local ecological conditions and their water need mountainous regions, people build ***diversion channels*** li the Western Himalayas for agriculture. ***Roof top rain wa*** is commonly practiced to store drinking water in Ra underground tanks are built inside the main house or cou connected to the sloping roofs of the houses through a p falling on the rooftops flows down into the tanks through technique is being intensively used in villages of Kerala In Meghalaya, a very old system of tapping stream and using bamboo pipes which is known as ***Bamboo drip irrig*** prevalent. About twenty liters of water enters the bamboo gets transported over hundreds of meters and finally red fifty drops per minute at the site of irrigation.

4.4.5 Floods

Flood is a general or temporary condition of partial or com of normally dry land areas from overflow of inland or tida the unusual and rapid accumulation or runoff of surface source. ***Flooding and flash flooding are the deadl disasters***. Floodwaters claim thousands of lives every millions homeless. One of the more frightening things that it can occur nearly anywhere, at any time. It can re water jams on rivers, even moderate rain, or a single very

Heavy downpour in the form of rain brings down more w disposed off by combined factors natural and man-made flooding. The rivers overflow embankments may be bre rains following storm and hurricane are heavy and brin amount of water causing floods.

In India, floods bring much havoc causing loss of life year. Due to flood, the plains become silted with mud and s the cultivable land areas. The worst suffering states Orissa, U.P. and West Bengal.

the water. An example is using wastewater to raise fish and grow non edible plants. Both activities remove nutrients in wastewater from food manufacturing operations.

- ***Renew used water***—Renewing wastewater is helping nature do its job. It may involve filtering to remove solid materials. In holding reservoirs, it might include promoting the growth of microbes so processes occur, such as the nitrogen cycle.
- ***Efficient use of water***—Everyone can make better use of water by consuming a little less water for the daily chores.

Water Harvesting for Conservation

In India, from the ancient times there exists an extraordinary tradition of water-harvesting system. People have in-depth knowledge of rainfall regimes and soil types and thus have developed various techniques to harvest rain water, ground water, river water and flood water in keeping with the local ecological conditions and their water needs. In hills and mountainous regions, people build ***diversion channels*** like the 'kuls' of the Western Himalayas for agriculture. ***Roof top rain water harvesting*** is commonly practiced to store drinking water in Rajasthan. Huge underground tanks are built inside the main house or courtyard and are connected to the sloping roofs of the houses through a pipe. Rain water falling on the rooftops flows down into the tanks through the pipes. This technique is being intensively used in villages of Kerala and Karnataka. In Meghalaya, a very old system of tapping stream and spring water by using bamboo pipes which is known as ***Bamboo drip irrigation system*** is prevalent. About twenty liters of water enters the bamboo pipe system and gets transported over hundreds of meters and finally reduces to forty to fifty drops per minute at the site of irrigation.

4.4.5 Floods

Flood is a general or temporary condition of partial or complete inundation of normally dry land areas from overflow of inland or tidal waters or from the unusual and rapid accumulation or runoff of surface waters from any source. ***Flooding and flash flooding are the deadliest of natural disasters***. Floodwaters claim thousands of lives every year and render millions homeless. One of the more frightening things about flooding is that it can occur nearly anywhere, at any time. It can result from excess water jams on rivers, even moderate rain, or a single very heavy downpour.

Heavy downpour in the form of rain brings down more water than can be disposed off by combined factors natural and man-made systems causing flooding. The rivers overflow embankments may be breached. Generally rains following storm and hurricane are heavy and bring unmanageable amount of water causing floods.

In India, floods bring much havoc causing loss of life and property each year. Due to flood, the plains become silted with mud and sand, thus affecting the cultivable land areas. The worst suffering states are Assam, Bihar, Orissa, U.P. and West Bengal.

Effects of Flood

(i) **Damage to buildings and other constructions:** Buildings and houses are washed away due to the impact of water under high stream velocity. They get destroyed and dislocated so severely that their reconstruction is not feasible. Sometimes damage is caused by inundation of buildings which means that the foundation remains intact while the damage to materials may be severe. In many cases the velocity of water may scour and erode the foundation of the buildings and other structures.

(ii) **Health effects:** The large amount of pooled water remaining after the floods leads to an increase in mosquito populations. Mosquitoes are most active at sunrise and sunset. People are exposed to malaria, dengue etc.

Swiftly moving shallow water can be deadly, and even shallow standing water can be dangerous for small children. Cars or other vehicles do not provide adequate protection from flood waters. Cars can be swept away or may breakdown in moving water.

(iii) Many wild animals are forced from their natural habitats by flooding, and many domestic animals are also without homes after the flood. General public is exposed to rabies. Animals are disoriented and displaced too. Rats may be problem during and after a flood. Flood waters can bury or move hazardous chemical containers of solvents or other industrial chemicals from their normal storage places. This can lead to accidents due to leakage or damage to the containers.

Flood Management

Flood forecasting has been recognized as one of the most important, reliable and cost-effective non-structural measures for flood management. Recognizing the crucial role it can play, Central Water Commission, Ministry of Water Resources has set up a network of forecasting stations covering all important flood prone interstate rivers. The forecasts issued by these stations are used to alert the Public and to enable the administrative and engineering agencies of the States/UT's to take appropriate measures.

Central Water Commission started flood-forecasting services in 1958 with the setting up of its first forecasting station on Yamuna at Delhi Railway Bridge. At present Central Water Commission has network of 159 flood forecasting stations.

4.4.6 Drought

Drought may be defined as an extended period - a season, a year or more - of ***deficient rainfall*** relative to the statistical multi-year average for a region. It is a normal and recurrent feature of climate and may occur anywhere in the world, in all climatic zones. Its features or characteristics, of course, vary from region to region.

In simple words, drought is a period of ***drier-than-normal conditions*** that lead to water-related problems. When rainfall is below normal for weeks, months or even years, it brings about a decline in the flow of rivers and streams and a drop in water levels in reservoirs and wells. If dry weather persists and water supply-related problems increase, the dry period can be called a 'drought'.

The major drought years in India were 1877, 1899, 1918, 1972, 1987 and 2002. Large parts of the country perennially reel under recurring drought. Over 68% of India is vulnerable to drought. The 'chronically drought-prone areas' - around 33% - receive less than 750 mm of rainfall, while 35%, classified as 'drought-prone' receive rainfall of 750-1,125 mm. The drought-prone areas of the country are confined to peninsular and western India - primarily arid, semi-arid and sub-humid regions.

Drought Classification

There are a number of classifications for drought. A **permanent** drought is characterised by extremely dry climate, drought vegetation and agriculture that is possible only by irrigation; **seasonal** drought requires crop durations to be synchronised with the rainy season; **contingent** drought is of irregular occurrence; and **invisible** drought occurs even when there is frequent rainfall, in humid regions.

Physical aspects are also used to classify drought. They may be classified into three major groups:

Meteorological drought is related to deficiencies in rainfall compared to the average mean annual rainfall in an area. There is, however, no consensus on the threshold of deficit that makes a dry spell an official drought. According to the India Meteorological Department (IMD), meteorological drought occurs when the seasonal rainfall received over an area is less than 75% of its long-term average value. If the rainfall deficit is between 26-50%, the drought is classified as 'moderate', and 'severe' if the deficit exceeds 50%.

Agricultural drought occurs when there is insufficient soil moisture to meet the needs of a particular crop at a particular point in time. Deficit rainfall over cropped areas during their growth cycle can destroy crops or lead to poor crop yields. ***Agricultural drought is typically witnessed after a meteorological drought, but before a hydrological drought.***

Hydrological drought is a deficiency in surface and sub-surface water supply. It is measured as stream flows and also as lake, reservoir and groundwater levels.

A sequence of impacts may be witnessed during the progression of a drought from meteorological, agricultural to hydrological. When drought begins, the agricultural sector is usually the first to be affected because of its heavy dependence on stored soil water. Soil water can deplete rapidly during extended dry periods.

If precipitation deficiencies persist, then people dependent on other sources of water begin to feel the effects of the shortage. Those who rely on groundwater, for instance, are usually the last to be affected.

When the situation returns to normal, and meteorological drought conditions have abated, the 'recovery cycle' follows the same sequence. Soil water reserves are replenished first, followed by stream flows, reservoirs/lakes and groundwater.

Causes of Drought

The principal cause of drought may be attributed to the ***erratic behaviour of the monsoon.*** The southwest monsoon, or 'summer monsoon' as it is called, has full control on agriculture, the Indian economy and consequently, the livelihoods of a vast majority of the rural population. The southwest monsoon denotes the rainfall received between the months of June and September and accounts for around 74% of the country's rainfall.

The coastal areas of peninsular India also receive rain from October to December due to periodic cyclonic disturbances in the Bay of Bengal (the north-east monsoon, or post-monsoon system)

Most parts of the peninsular, central and north west India, the regions most prone to periodic drought receive less than 1,000 mm of rainfall. There are other reasons, mostly man made, which aggravate drought or create drought like situations in the country. India is well-endowed in terms of rainfall, with Cherrapunji receiving an annual rainfall of around 11,000 mm. Even Saurashtra and the Kutchh region record rainfall of around 578 mm. India's average rainfall is around 11,70 mm, yet the country suffers recurrent drought.

The ***over-exploitation of surface and groundwater*** is another major cause of water stress leading to drought situations in the country.

India has seen a sharp decline in groundwater levels, leading to a fall in supply, saline water encroachment and the drying of springs and shallow aquifers. Around 50% of the total irrigated area in the country is now dependent on groundwater, and 60% of irrigated food production depends on irrigation from groundwater wells.

In some regions of Gujarat, Rajasthan, Tamil Nadu, Karnataka, Punjab and Haryana the decline in water levels due to over exploitation has been to an extent of 1 - 2 meters/yr.

The ***rapid depletion of forest cover*** is also seen as one of the reasons for water stress and drought. India has 22% of its total geographical area under forest cover, which is much lower than the prescribed global norm of 33%. Although the scientific evidence is inadequate, forest-water linkages are widely acknowledged, especially the watershed functions of forests, greater availability of water, less soil erosion, more rainfall, flood and landslide control, etc.

Impact of Drought

Drought has a direct and indirect impact on the economic, social and environmental issues of the country. Depending on its reach and scale it could bring about social unrest.

The immediate visible impact of monsoon failure leading to drought is felt by the agricultural sector.

- The surface water and the ground water levels drop.
- The food grain production goes down.

Water and fodder shortages during drought situation cause considerable stress on the bovine population also.

The impact of droughts can be reduced through mitigation and preparedness. Planning well in advance to mitigate drought could relieve the most suffering and at lesser expense.

4.4.7 Conflicts Over River Water

Water availability in different parts of a basin, and the linkage between land use and water use entails that activities of one kind in one part of the river basin can negatively or positively influence other activities in different parts of the basin. The failure to perceive this inter-relation in the planning of water utilisation has become a major source of conflict over water use in river basins.

Water allocation between conflicting demands for water have rarely taken full account of the underlying conflicts and have therefore aggravated the problems of inequalities and maldistribution. The four major categories of use on which water planning is based are:

1. Domestic
2. Agricultural
3. Industrial
4. Power generation

Rarely the internal conflicts in each sector or between sectors are clearly defined in water development projects. On the contrary it has been assumed, for instance, that multi-purpose river valley projects that provide irrigation as well as generate hydro-power do not have conflicting uses. However, the very location of these projects is primarily determined on the basis of either of these objectives and water releases are also determined by priorities for power or for irrigation. Other inter-sectoral conflicts include diversion of water for irrigation from drinking water, or for industry from agricultural and domestic use. Not only do diverse uses conflict with each other inter-sectorally, they can also conflict intra-sectorally on the basis of conflicting interests between the rich and powerful and the poor and marginal. The category 'domestic' as an undifferentiated one conceals the conflict between the poor rural peasants requiring a pot full of drinking water and the rich urban elite using large quantities of water for meeting the requirements of water-intensive sewage systems, space cooling, gardening, etc. Domestic requirements vary for different people and the high demands from urban areas are often met by diverting water from rural areas. Similarly, an undifferentiated category of agriculture conceals conflicts between water-intensive cultivation of commercial crops for high cash returns and prudent water use for protective irrigation of staple food crops essential for survival.

Social conflicts over water can also be analysed at different societal levels. Thus, inter-state conflicts are generated when water projects of upstream states influence the quality and quantity of water flow in the basin and reduce the possibilities of water use by downstream states. Major inter-basin water transfers in rivers flowing through many states also generate conflicts by disturbing the rights of states. Conflicts also arise between the state and the people when official planning and policies lead to changes in water use and utilisation pattern and therefore undermine people's access to water. Thus, state planned quarrying of minerals or timber extraction in the river catchments affect the river flow and generate conflicts downstream. Similarly, state planned agricultural production based on large irrigation projects to generate marketable surpluses of cash crops conflicts with people's needs for local food production. Such projects also lead to conflicts between the state and the people by eroding traditional water rights which are often communal in nature and ensure the survival of all members of the community. Local common management of water resources and the ethics and values on which it is based are frequently modified by government planned and managed water projects aimed at the expansion of commercial agriculture. Finally, state plans tend to serve the interests of the economically and politically powerful groups of society and hence generate new gaps between the rich and poor in terms of access to water resources.

4.4.8 Case Study

CASE STUDY

Kaveri River Water Dispute

The water of the river Kaveri has been the cause of a serious conflict between Karnataka and the state of Tamil Nadu. Over the years, the dispute has become increasingly complex due to the stubborn attitude of the parties involved, particularly those of the states of Karnataka and Tamil Nadu. The dispute at its root is a question of the sharing of the waters of the river Kaveri. While the state of Tamil Nadu has historically enjoyed a vastly greater usage of the waters compared to Karnataka. Karnataka on the other hand, sees it as a grave historic injustice that has been forced upon it. The origin of this disparity itself, lies in a two controversial agreements signed in 1892 and then in 1924 between the Madras Presidency and the Princely State of Mysore.

Karnataka claims that these agreements were in favour of the Madras Presidency. It claims that these agreements dealt its own interests and therefore wants a renegotiated settlement based on equitable sharing of the water. Tamil Nadu on the other hand, pleads that it has already developed almost 3,000,000 acres (12,000 km^2) of land and as a result has come to depend very heavily on the existing pattern of usage. Any change in this pattern, will adversely affect the livelihood of millions of farmers in the state.

Decades of negotiations between the parties involved bore no fruit and the Government of India finally constituted a tribunal in 1990 to look into the matter. The tribunal after hearing arguments of all the parties involved for the last 16 years, delivered its final verdict on February 5, 2007. In its verdict, the tribunal allocated 419 billion ft^3 ($12\ km^3$) of water annually to Tamil Nadu and 270 billion ft^3 ($7.6\ km^3$) to Karnataka; 30 billion ft^3 ($0.8\ km^3$) of Kaveri river water to Kerala and 7 billion ft^3 ($0.2\ km^3$) to Puducherry. The dispute however, seems far from over with all four states deciding to file review petitions seeking clarifications and possible renegotiation of the order.

4.4.9 Dams-benefits and Problems

India is one of the largest dam building nations in the world. There are almost 4291 dams in India. 3596 have been built and 695 are under construction (World Commission on Dams, Report 2000). Dams offer huge benefits in terms of electrical power generation and irrigation.

The various benefits of dams are:

(i) *Hydroelectric power*– Electrical power generation is a major benefit which may be cheaper and environmentally safer than energy.

(ii) *Irrigation*– Regions with poor or unpredictable rainfall can be turned into fertile farmland.

(iii) *Water supply*– Provide dependable water supply for urban and industrial use.

(iv) *Flood control*– Holding back and channeling potentially dangerous water flow.

(v) *Navigation*– Multipurpose river valley projects also provide for inland water navigation.

Some problem associated with dams are:

(i) *Environmental*– Dams may destroy wild life habitats, drain wetlands and cause river pollution by reducing the river flow to a level where the river can no longer self-cleanse. Farmland can be ruined by salt produced by the irrigation process.

(ii) *Risk of failure*– Sometimes dam fails due to landslides or earthquakes which can be catastrophic.

(iii) *Submergence*– of fertile fields and human settlements.

(iv) Dams have serious consequences with regards to the *displacement of tribal people*. This displacement results in transfer of resources from weaker sections of the society to the more privileged sections.

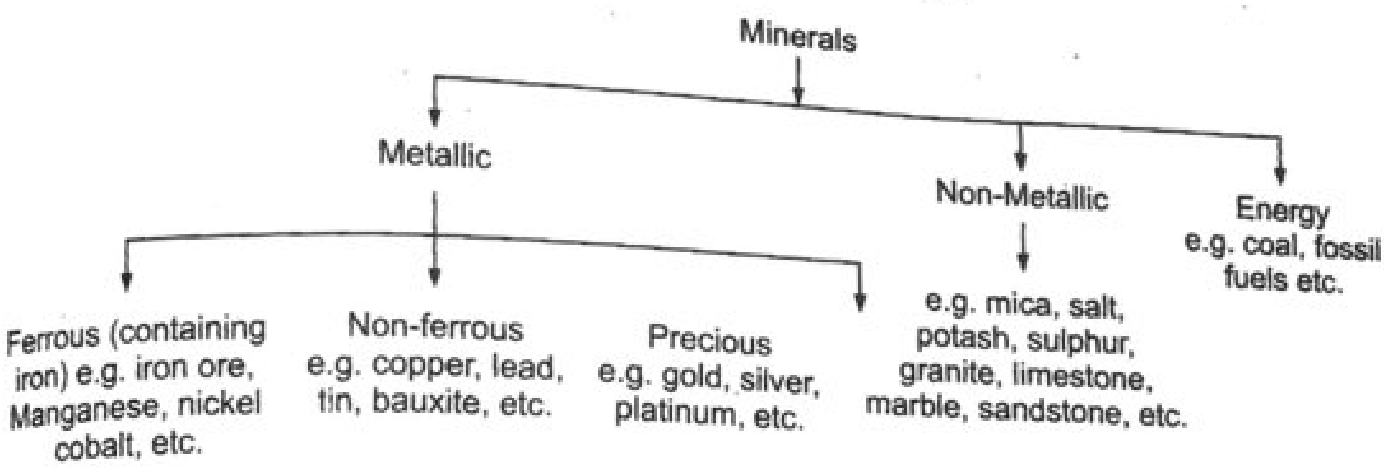
(v) *Cost*– Dams are very expensive to build and may not provide sufficiently economical electrical power generation, water supply for irrigation.

4.5 MINERAL RESOURCES

India is rich in mineral resources. It has fairly abundant reserves of iron ore and mica and adequate supplies of manganese ore, titanium, bauxite and coal. There is a deficiency of copper, lead, zinc and gold. The country earns foreign exchange from the export of minerals like iron ore, titanium, manganese ore, bauxite and granite. In all there are over 3000 mines in India. About 8 lakh people are employed in the mining sector. It accounts for about 11% of the country's industrial output. Industrial development of the country depends upon this sector.

4.5.1 Classification of Minerals

Geologists classify minerals according to their chemical composition and crystalline structure. Generally minerals are classified into three categories metallic, non-metallic and energy. Metallic minerals are subdivided into ferrous (containing iron), non-ferrous (containing metals other than iron) and precious (Table given below):



Minerals occur in different types of rocks. Some are found in igneous rocks, while others in sedimentary rocks. Almost all metals are found in the form of ore. The ore contains impurities and therefore processing is required before use.

4.5.2 Distribution of Minerals and their use

Iron Ore

It is the back bone of modern civilization. It is used for manufacturing of machines, agricultural implements and items of general use. India shares about 20% of the world reserve of iron ore. Most of the mines in the country are in Chhattisgarh, Jharkhand, Orissa, Goa and Karnataka. About half of the iron ore produced in the country is exported primarily to Japan, Korea, European countries and gulf countries.

Manganese Ore

It is used for making iron and steel and preparing alloys. It is also used to manufacture bleaching powder, insecticides, paints and batteries. About one-fifth of the world's deposits of manganese ore are found in India. The country stands second in the reserve of manganese ore in the world after

Zimbabwe. The main reserves are in Karnataka, Orissa, Madhya Pradesh, Maharashtra and Goa and Madhya Pradesh produces more than half of the national total.

Copper

It is used for making utensils, electric wire and alloys. Indian copper ore contains less than 1% of copper (international average is 2.5%) and hence its mining and smelting is costly affair. About 90% of the reserves are concentrated in Madhya Pradesh, Rajasthan, Jharkhand, Karnataka and Andhra Pradesh, Production of copper in the country is less than the requirement, hence it is imported.

Lead

The ore of lead is known as galena. It is a soft and heavy metal and is a bad conductor of heat. It is used in cable covers, ammunitions, paints, glass and rubber making. Lead ore occur in Rajasthan, Andhra Pradesh and Tamil Nadu. India produces about 25% of its requirement and thus rest is imported from Australia, Canada and Myanmar.

Bauxite

It is an ore from which aluminium is extracted. Aluminium is a light metal used in the manufacturing of aeroplanes, utensils and other household goods. India has vast reserve of bauxite and occurs mainly in Jharkhand, Orissa, Gujarat, Maharashtra, Chhattisgarh, Madhya Pradesh and Tamil Nadu.

Mica

It has insulating properties and can withstand high voltage, hence its major use is in electrical and electronic industries. Mica reserves occur in Jharkhand, Bihar, Andhra Pradesh and Rajasthan. India produces about 60% of world's production of mica.

Limestone

It is found associated with rocks composed of calcium carbonates or calcium or magnesium carbonates. Major use (about 75%) of limestone is in cement industry; rest is used for smelting of iron and in chemical industries. Almost all states have some quantities of limestone, but about three fourths of the country's production comes from Madhya Pradesh, Chhattisgarh, Andhra Pradesh, Rajasthan, Gujarat, Karnataka and Himachal Pradesh.

4.5.3 Exploitation of the Mineral Wealth

Minerals often require quite a lot of processing to get the desired metals from them. The steps involved are:

(i) locating a supply of mineral
(ii) mining
(iii) processing the mineral to get desired products.

Locating Minerals

Minerals could be sometimes located below the surface of the Earth's crust. They may be mixed with other materials that have almost no value. Thus geologists study the surface of the earth carefully and prepare maps showing mineral deposits. On-site inspections alongwith aerial photography, remote sensing and global positioning systems gives additional information for the exploration and finding new locations where mineral deposits could be found. Drilling may be used to collect samples below the surface. The ocean water contains vast quantitites of mineral, but most of these are too widely diffused to be of economic significance. However, common salt, magnesium and bromine are largely derived from ocean waters. The ocean beds, too, are rich in manganese nodules.

Mining

Depending upon the location of the ores, the following methods are adopted for their extraction:

- ***(i) Surface mining***
- ***(ii) Sub-surface mining***

Surface Mining

This is adopted when the mineral is above or just below the surface. The layers of rock and soil covering the mineral deposits are first scrapped off and discarded as spoils. The minerals are then removed using appropriate technology. The type of surface mining employed depends upon the mineral and the topography of the mine. Thus different methods are adopted in surface mining:

Open-pit and Strip Mining Open-pit surface mining is digging a large hole in the earth to get the minerals. The large pit is a *quarry*. Granite iron and copper ore and sand are often mined with open pits.

Strip mining is digging a long trench to get the minerals. A series of strips are dug. The strips are parallel to each other. The soil and rock that are not extracted are put into the previous strip. Rows of dug-out materials, known as spoil banks, are created on the surface.

Subsurface Mining

A subsurface mine is deep inside the earth to remove minerals such as coal and valuable minerals like gold ore. The surface of earth may be disturbed very little. A mass of tunnels or shafts may be used to go into the deposits of minerals. Thus these mines are also termed as *deep shaft mines.* This type of mining is more expensive and complex than surface mining. The shafts may be dug straight down to the mineral ore or they may slope downward. The tunnels and benches are formed inside the earth so that the ore can be extracted and transported out. Although the environment is not visibly harmed, there is always the risk of roof collapse, explosion of gases, and disturbance to underground water movement.

Processing the Mineral

Ore needs to be separated from material that does not contain the desired metal. The processing needed varies with the mineral and its use. Common processing procedures include:

- **Grinding and crushing:** Grinding and crushing are used to set materials into the desired sizes. Grading may be used to separate the materials based on size.
- **Sorting:** Various ways may be used to separate valuable metals from the ore. Water may be used to 'float' metal materials from the ore. One function of sorting is to separate metal from ore.
- **Smelting:** Smelting is heating ore so the metal separates from the undesirable material. The process may also be used to cast the metal into the desired products. Large furnaces are used in smelting. Most smelting produces several products that can be used. Iron smelting usually produces silicon and phosphorus. Copper smelting may produce sulfur and other materials. Both iron and copper smelting produce slag. Since iron melts at 1,535°C, iron ore may be heated to 1,600°C. The iron is molten or liquid-like. The melting points of other metals vary. In liquid form, the metal separates from the gases and solid materials. The metals with the lowest melting points become liquid earlier and can be removed.
- **Purification:** Various methods are used to purify the metal from the ore. Smelting often involves some purification.

Some minerals like limestone, sand, gravel and kaolin require less processing. No smelting or purification is needed.

After processing a lot of solid waste materials, known as Tailings, remain depending upon the yield of ore. For e.g. copper ore produces only about 1% copper. This means that a ton of copper ore will result in nearly a ton of tailings.

4.5.4 Conservation of Minerals

The human beings have a strong dependence upon mineral deposits for the growth and development of the nation. In this process the mineral resources are being rapidly consumed. The geological processes of mineral formation are so slow that the rates of replenishment are infinitely small in comparison to the present rates of consumption. Mineral deposits of the country are thus, valuable but shortlived possessions. A serious effort has to be made in order to use the mineral resources in a planned and sustainable manner. Improved technologies need to be constantly evolved to allow use of low grade ores at low costs. Recycling of metallic scrap like steel, copper, aluminium, zinc, lead etc. should be encouraged and facilitated by fixation of appropriate standards for classification and grading of scrap and adoption of fiscal measures. Similarly utilisation of low grade minerals, mineral waste and rejects should also be encouraged through appropriate incentives.

4.5.5 Environment Impact of Mineral Use

Locating minerals, mining and processing minerals affects the environment. There is no way to get and use minerals without some intrusion into the environment. Using good practices helps keep damage to the environment to a small extent.

Mining Damage: Mining can disrupt the earth's natural landscape and forest ecosystems. Forests are destroyed and wild animals are driven away. Several species may become extinct. Excavations makes the surface unsightly causing land degradation. The top soil may be removed and large areas excavated to expose minerals. These areas may collect water and create polluted run-off. Sometimes these areas become breeding ground for mosquitoes. Large piles of tailings are sometimes left. Excavated land can erode and damage water quality. Mined areas may have substances that react with water to form acid. Water that seeps or runs out of mine may contain acid. For example, iron ore often has sulfur mixed in it. Runoff from the mine will have a high acid content. Streams and lakes may be damaged by the acid runoff. With other minerals toxic materials like lead from lead mine may enter the runoff.

The land areas damaged by mining can be filled and graded to a natural contour. Grasses and trees may be planted on the land. Often little top soil is available for plants. Some mine sites may contain toxic substances that kill common vegetation. Special kinds of plants may be needed to grow in these places.

Waste Disposal: Tremendous amounts of solid waste materials may result from processing minerals. Often 80 percent or more of the ore may be waste after processing. The amount of tailing produced is greater with low-grade ore. Large piles of tailings may be seen at smelting plants. The tailings are sometimes used as fill for low land areas.

Some solid waste products from smelting are put to good use. Slag is used as a source of phosphorus and calcium in fertilizer. Slag is also used in road paving and making lighter-weight concrete.

Emissions Control: Smoke, steam, particulates, and other materials may be released by mining. Some of these cause serious pollution problems.

Burning coal to heat minerals in smelting releases substances into the air. The minerals themselves also give off gases in smelting. Sulfur, nitrogen, and carbon are commonly released causing air pollution. These materials also find their way into the water and soil. It may result to acid rain, unproductive soil, and water without living organisms.

The Clean Air Act and Clean Water Act regulate air and water pollution. Mining and processing are covered in these Acts from the standpoint of what can be released into the air and water.

4.5.6 Case Study

CASE STUDY

No Mining in Aravalis

In a deed to preserve the ecology of sensitive Aravali hill, the Supreme Court rejected pleas for lifting the ban on mining activities in the range. The court had for the first time on May 6, 2002 banned mining activities in the Aravali hill on noticing that several illegal mining activities in the range were harming the range putting immense pressure on the ecology of the area.

The Court has now constituted a high-level Monitoring Committee to suggest ways and means for its "overall ecological restoration". Declaring that "ecology of the Aravali Hills has to be preserved at any cost", the Court directed the Committee to go into all issues pertaining to mining in the area and suggested remedial measures for eco-restoration. The court said that the Committee, apart from suggesting methods for "overall ecological restoration", would monitor implementation of several reports submitted previously by various other committees. The committee would also inspect and assess the mines that were operational till the ban orders came into force in May 2002 and recommended whether these could resume operation on the basis of "sustainable development principle", it said. The Court, however, made it clear that each mine owner, before applying for renewal of mining lease, should have to obtain environmental assessment clearance from the concerned authorities.

Attaching a premium to green areas, the apex Court said that under no circumstances mining could be allowed in areas where afforestation efforts have been undertaken. In the same breadth, Court said that if the Committee found that the legally permitted mining going on in the area was " adversely affecting the ecology of the hill range", then it could consider suggestions for an absolute ban on these activities in the area.

Aravali hills, spanning three states of Delhi, Haryana and Rajasthan, stand as a natural barrier to the Thar desert from extending into the plains.

4.6 FOOD RESOURCES

The basic necessity for the survival of human beings is food. The food humans eat are composed of several major types of biological molecules necessary to maintain health: carbohydrates, proteins and lipids. In addition to these, humans require minerals, vitamins and water in their diet.

The three main food sources are:

- Croplands that provide 76% of the total foodgrains.
- Rangelands that produce meat and milk from grazing livestock, accounting for 17% of total food.
- Fisheries that supply the remaining 7%.

Food production is probably the most important natural resource issue facing the world today. Population growth has necessitated a continuing expansion of agricultural output. The expansion must take place in the context of severe constraints on the availability of agricultural land and on the ability of natural biogeochemical cycling processes to supply agricultural inputs such as water and absorb waste products such as nitrogen. Thus we have to achieve almost a double agricultural output over the next half century through more intensive use of existing lands while minimising the environmental impacts of this intensification.

4.6.1 World Food Problems

The U.N. Food and Agricultural Organisation (FAO) in 2004 estimated that more than eight hundred million people lack access to the food needed for healthy, productive lives. Most of these people live in the rural areas of the poorest developing countries. The two region of the world with the greatest food insecurity are South Asia, with an estimated two hundred seventy million hungry people, and Sub-Saharan Africa, with an estimated one hundred and seventy five million (Sub-Saharan Africa includes all African countries located south of the Sahara Desert).

The average human must consume enough food to get approximately 2600 Kcal per day. (The average man requires 3000 Kcal per day whereas the average woman requires 2200 Kcal per day). If a person consumes less than the required amount over an extended period, his or her health and stamina decline, even to the point of death. People who receive fewer calories than needed are ***undernourished.*** Worldwide, one hundred and eighty two million children under the age of five suffer from ***undernutrition*** and are seriously underweight for their age, according to the World Heath Organization (WHO). This accounts to almost one-third of all children under five in developing countries.

The total number of calories consumed is not the only measure of good nutrition. People can receive enough calories in their diets but still be ***malnourished*** because they are not receiving enough of specific, essential nutrients such as proteins, vitamin A, iodine or iron. For example, a person whose primary food is rice can obtain enough calories, but a diet of rice lacks sufficient amounts of proteins, lipids, minerals, and vitamins to maintain normal body functions. Adults suffering from ***malnutrition*** are more susceptible to disease and have less strength to function productively than those who are well fed. In addition to poor physical development and increased disease susceptibility, children who are ***malnourished*** do not grow or develop normally. Because ***malnutrition*** affects cognitive development, ***malnourished*** children do not perform as well in school as children who are well fed. Currently, WHO estimates that more than 3 billion people worldwide—the greatest number in history—are malnourished. In addition, more than half the deaths in children less than 5 years old in developing countries are associated with ***malnutrition.***

The two most common diseases of malnutrition are **marasmus** and **kwashiorkor**. Marasmus (from the Greek word **marasmos,** meaning 'a washing away') is progressive deterioration caused by a **diet low in both total calories and protein**. Marasmus is most common among children in their first year of life—particularly children of poor families in developing **nations.** Symptoms include a pronounced slowing of growth and extreme

Shifting cultivation supports relatively small populations. ... agriculture is one of several distinct types of shifting c... involves clearing small patches of tropical forests to plant cr... tropical soils lose their productivity quickly when they ... farmers using slash-and-burn agriculture must move from ... another every three years or so, thus this agriculture is la...

Nomadic herding in which livestock is supported by land ... successful crop growth, is another type of land-intensive ... agriculture.

Intercropping is another form of intensive subsistence agri... involves growing a variety of plants simultaneously on same fie...

4.6.4 The Green Revolution

By the middle of the 20th century, serious food shortages occur... developing countries including India. It was widely recognized that ... food supplies were needed to feed their growing populations. The ... and introduction during the 1960's of high yielding varieties of whe... gave the chance to provide the people with adequate supplies of foo... yielding varieties required intensive cultivation methods, including ... of commercial inorganic fertilizers, pesticides and mechanized ... to realise their potential. These agricultural technologies were pass... highly developed nations to developing nations.

Using modern cultivation methods and the high-yielding var... certain staple crops to produce more food per acre of cropland is ... the ***green revolution.*** The two most important problems associa... higher crop production are the high energy costs built into this ... agriculture alongwith the serious environment problems caused ... use of the commercial inorganic fertilizers and pesticides.

4.6.5 Effects of Modern Agriculture

The practices of industrialised agriculture have resulted in ... environmental problems that impair the ability of non-agric... terrestrial and aquatic ecosystems to provide essential ecosystem ...

The agricultural use of fossil fuels and pesticides produces air p... Untreated animal wastes and agricultural chemicals such as ... and pesticides cause water pollution that reduces biological divers... fisheries etc. The agricultural practices are one of the major ... surface water pollution in India. Nitrates from animal wa... commercial inorganic fertilizers are probably the most ... groundwater contaminant in agricultural areas.

Fertilizers and Pesticides

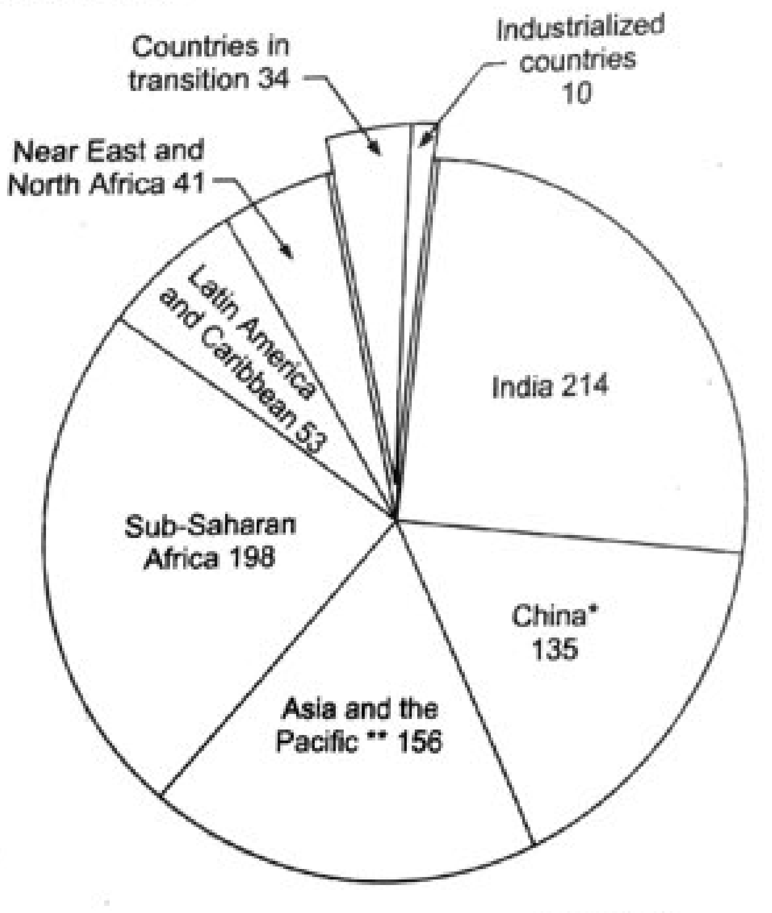
Average yields of grains worldwide more than doubled between 1960 ... owing mostly to increases in the use of *fertilizers*. Fertilizer use in... four times in that same 30-year period. One of the main reaso... prominence is the development of high-yielding plant varieties ...

atrophy (wasting) of muscles. It is possible to reverse the effect of marasmus with an adequate diet.

Kwashiorkor (a native word in Ghana meaning 'displaced child') is malnutrition resulting from **protein deficiency**. It is common among children in all poor areas of the world. The main symptoms include edema (fluid retention and swelling); dry, brittle hair; apathy; stunted growth; and sometimes mental retardation. One of the most typical features of kwashiorkor is a pronounced swelling of abdomen. Kwashiorkor is treated by gradually restoring a balanced diet.

People who eat food in excess of that required are ***overnourished***. Generally, a person suffering from ***overnutrition*** has a diet high in saturated (animal) fats, sugar, and salt. Overnutrition results in obesity, high blood pressure, and an increased likelihood of such disorders as diabetes and heart disease. In nutrition experiments, ***overnutrition*** in rodents resulted in a higher incidence of cancer compared with rodents fed a calorie-restricted diet, but evidence that links overnutrition to cancer in humans is sparse. Many human studies show a correlation between diets high in animal fat and red meat and certain kinds of cancer (colon and prostate). ***Overnutrition*** is most common among people in highly developed nations, such as the United States, where the Pan American Health Organization estimates that two out of three adults are overweight, and nearly one in three is obese. ***Overnutrition*** is also emerging in some developing countries, particularly in urban areas. As people in developing countries earn more money, their diets shift from primarily cereal grains to more processed foods and livestock products.

The figure below shows the scale of undernourishment, 1999–2001 (millions).



* includes Taiwan province of China

** exc. China and India

Fig. 4.3

4.6.2 Impacts Caused by Agriculture

Agricultural activities represent an enormous transformation of natural ecosystems occurring over very large portions of the Earth's surface. As such, they inevitably cause vast environmental impacts. These impacts could be broadly classified into:

- On-site impacts
- Off-site impacts

On-site impacts include soil erosion and range land degradation.

Soil erosion on agricultural lands takes place through three major processes; overland flow (or run off), wind and streambank erosion. Of these processes, overland flow erosion is the most visible and widespread, and in most agricultural areas it is quantitatively the most important. Wind erosion occurs on exposed soil if strong winds blow at times when the soil surface is relatively dry. Streambank erosion is limited to fields that border streams, and though locally significant it is not a major factor in soil erosion worldwide.

Rangeland Degradation Range, or grazing land, provides forage for limited numbers of domestic animals. *Overgrazing* occurs when the number of animals on these lands exceeds carrying capacities. Several areas of the world, notably the dry lands around the Mediterranean Sea, have been long overgrazed, with resulting problems of devegetation, erosion, and ultimately the threat of desertification. *Desertification* is land degradation in dryland regions resulting mainly from adverse human impact. It occurs in parts of all the major semiarid regions of the world and affects some of these regions more than others.

The **off-site impacts** include water pollution from agricultural run-off, air pollution from blowing dust or dispersal of agricultural chemicals like fertilizers and pesticides.

4.6.3 Types of Agriculture

Agriculture can be roughly divided into two types:

- industrialized or modern
- subsistence agriculture

Most farmers in highly developed countries and some in developing countries practice ***industrialized agriculture*** or high-input agriculture. It relies on large inputs of capital and energy, in the form of fossil fuels, to produce and run machinery, irrigate crops and produce agrochemicals such as commercial inorganic fertilizers and pesticides. This type of agriculture produces high yields enabling forests and other natural areas to remain wild instead of being converted to agricultural land.

In the developing countries most farmers practice ***subsistence agriculture***, the production of enough food to feed oneself and one's family with little left over to sell or reserve for hard times. Subsistence agriculture, too, requires a large input of energy, but from humans and draft animals rather than from fossil fuels.

Shifting cultivation is a form of subsistence agriculture in which short periods of cultivation are followed by longer periods of fallow (land left uncultivated) in which the land reverts to forest.

Shifting cultivation supports relatively small populations. Slash-and-burn agriculture is one of several distinct types of shifting cultivation that involves clearing small patches of tropical forests to plant crops. Because tropical soils lose their productivity quickly when they are cultivated, farmers using slash-and-burn agriculture must move from one area to another every three years or so, thus this agriculture is land-intensive.

Nomadic herding in which livestock is supported by land too arid for successful crop growth, is another type of land-intensive subsistence agriculture.

Intercropping is another form of intensive subsistence agriculture that involves growing a variety of plants simultaneously on same field.

4.6.4 The Green Revolution

By the middle of the 20th century, serious food shortages occured in many developing countries including India. It was widely recognized that additional food supplies were needed to feed their growing populations. The development and introduction during the 1960's of high yielding varieties of wheat and rice gave the chance to provide the people with adequate supplies of food. But the yielding varieties required intensive cultivation methods, including the use of commercial inorganic fertilizers, pesticides and mechanized machinery, to realise their potential. These agricultural technologies were passed from highly developed nations to developing nations.

Using modern cultivation methods and the high-yielding varieties of certain staple crops to produce more food per acre of cropland is known as the ***green revolution.*** The two most important problems associated with higher crop production are the high energy costs built into this form of agriculture alongwith the serious environment problems caused by the use of the commercial inorganic fertilizers and pesticides.

4.6.5 Effects of Modern Agriculture

The practices of industrialised agriculture have resulted in several environmental problems that impair the ability of non-agricultural terrestrial and aquatic ecosystems to provide essential ecosystem services.

The agricultural use of fossil fuels and pesticides produces air pollution. Untreated animal wastes and agricultural chemicals such as fertilizers and pesticides cause water pollution that reduces biological diversity, harms fisheries etc. The agricultural practices are one of the major cause of surface water pollution in India. Nitrates from animal wastes and commercial inorganic fertilizers are probably the most widespread groundwater contaminant in agricultural areas.

Fertilizers and Pesticides

Average yields of grains worldwide more than doubled between 1960 and 1990, owing mostly to increases in the use of *fertilizers.* Fertilizer use increased over four times in that same 30-year period. One of the main reasons for this prominence is the development of high-yielding plant varieties that require large inputs of fertilizer to realize their potential. In addition, fertilizers may make production possible on otherwise marginal land.

The three most important nutrients required by plants are nitrogen, phosphorus, and potassium. Nitrogen is ultimately derived from the atmosphere, but it is made available to plants by nitrogen-fixing bacteria. It is the nutrient that is most often deficient and that is most widely applied to crops. Additions in amounts of 45 to 90 pounds/acre (50 to 100 kg/ha) may increase yields from 1.5 to 3 times, depending on plant variety and inherent soil fertility. Natural gas is an important raw material in the manufacture of most nitrogen fertilizers. The most commonly used forms are ammonia (NH_3) and urea ($CO[NH_2]_2$). Phosphorus is usually present in small quantities in soils, but it is often found in relatively unusable forms. It is usually applied as superphosphate or as phosphoric acid, which are manufactured from phosphate rock deposits. In soils, potassium generally is found in larger quantities than phosphorus because it is a more abundant constituent of most rocks. Plants also demand it in large quantities, and in many areas potassium fertilization is important. In some areas, local soil conditions or the particular needs of plants require that other fertilizers be added, with lime (a source of calcium and magnesium as well as a regulator of soil pH) being the most common.

Organic fertilizers (primarily manure) have historically been the most important source of nutrients, especially nitrogen. Organic fertilizers also help maintain good soil structure and water-holding capacity by keeping soil organic matter content high. In the wealthy nations, inorganic fertilizers are now more important, but in the developing nations manure is still a common fertilizer. In most areas, manure supplies are quite limited, and manure is more difficult to apply than other forms of fertilizer. Manure is low in nutrient content relative to synthetic fertilizer and is not capable of providing the large inputs of nutrients demanded by high-yielding crop varieties. Increasingly, therefore, inorganic sources of nutrients have been replacing organic sources.

Pesticides is a general term referring to any of the number of chemical agents used to control organisms harmful to plants, including insects, fungi, and some types of worms. Pesticides include insecticides, rodenticides, fungicides and others. Herbicides are used to control weeds. The use of pesticides and herbicides has accounted for a large part of recent increases in crop yields. Thousands of different kinds of pesticides and herbicides are in use, and the vast majority are complex organic compounds manufactured using petroleum as an important raw material. Among insecticides, organochlorines, organophosphates, and carbamates are important types.

The first widely used insecticides were organochlorines such as DDT, aldrin, dieldrin, and chlordane. In the 1960s and 1970s, these were largely replaced by organophosphates for most uses, in part because insects began to develop resistance to the effects of organochlorines and in part because orgnophosphates break down more rapidly and therefore are less likely to accumulate in the environment. Today many different types of chemicals and application methods are used. Among the small-grain crops, pesticides and herbicides are used most intensively on corn and soybeans. Most fruits and vegetables are susceptible to damage by insects, fungi, and other pests, and various pesticides are used depending on specific circumstances.

One of the consequences of indiscriminate use of pesticide is the adverse health impact on society in general and vulnerable population like children in particular. Some of the well-known health effects of pesticide exposure include acute poisoning, cancer, neurological effects, reproductive and developmental harm. The major causes of concern are bio-accumulation of pesticides and time period that it takes to express the negative health consequences.

Indian Scenario

After independence the use of fertilizers in India in the last 50 years has grown nearly 170 times. In 1950 use of fertilizer per hectare in India was 0.55 kg but by 2001-02 this figure has increased to around 90.12 kg per hectare. Green revolution during 1960s and subsequent increased intensification of agriculture were major causes behind this growth as seen in Fig. 4.4.

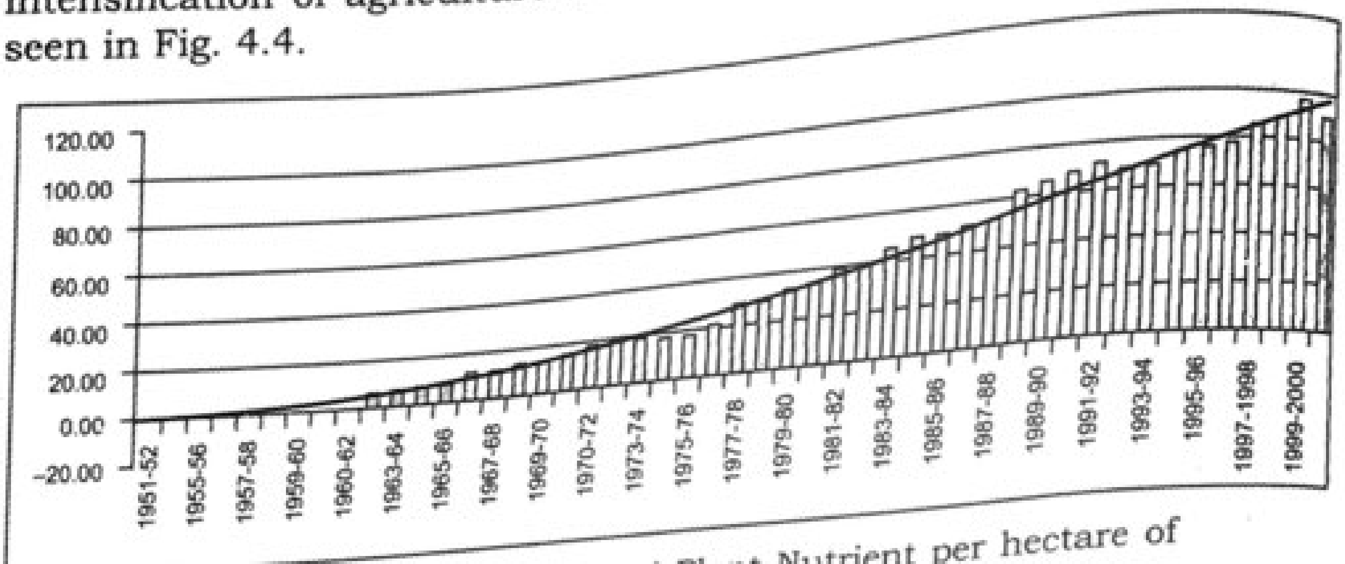


Fig. 4.4 Consumption of Total Plant Nutrient per hectare of Gross Cropped Area (in kg)

In India consumption of insecticide in agriculture has been increased more than 100% from 1971 to 1994-95. For instance, insecticide consumption in India, which was 22013 tonnes has increased to 51755 tonnes by 1994-95. Consumption of all of these pesticides in same duration has increased more than two times, that is from 24305 tonnes to 61357 tonnes (Fig. 4.5).

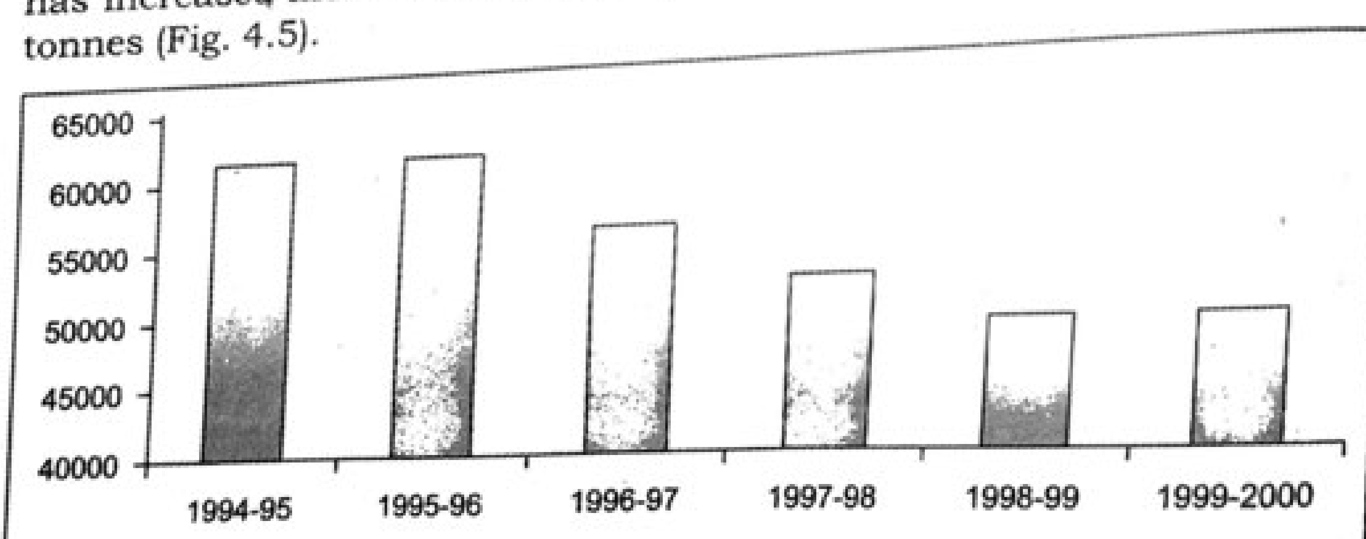


Fig. 4.5 Consumption of Pesticides in India (in tonnes)

But in recent past, change has been observed in trends of pesticides consumption. As a consequence of adoption of bio-intensive Integrated Pest Management Programme in various crops the consumption of chemical pesticide (Tech. Grade) has come down from 66.36 thousand MT during 1994-95 to 43.59 thousand MT during 2001-02 with a reduction of 27.69% (Thirty Seventh Report of Standing Committee on Petroleum and Chemicals 2002).

4.6.6 Salinity and Waterlogging

Salinity can be of natural occurance in drylands, arid area and irrigated rural areas where the concentration of soluble salts in the soil increases. This is mostly due to low rainfall, poor drainage and high temperatures, when water evaporates quickly leaving behind the highly concentrated salts.

Where human activity interferes with the natural process of 'movement of water', then it will directly contribute to salinity and water-logging by causing water-tables to rise especially in those areas where natural geological features and conditions make them more prone to rising water-tables

Salinity can be prevented or checked by

- Improving the drainage
- Reclamation of salinated lands by leaching with plenty of water.

Due to intensive agricultural management practices in Haryana and Punjab, many hectares of good agricultural land turns saline every year.

Waterlogging occurs when the soil-surface area becomes saturated and soil pores are full of water.

Water logging could occur due to

- Periods of heavy rain.
- Poor irrigation management
- Poor drainage
- Rising water table (as discussed above)

Water table could also rise due to overwatering with irrigation, because it is the unused or excess portion of water applied that recharges the ground water system causing the water table to rise.

Effects of water logging are

- Water-logged soil pores have no oxygen. Plants need oxygen to breathe and grow.
- Vegetation can turn yellow, growth is stunted and thin.
- Trees and plants can die and bare patches of soil appear.
- Plants species more tolerant of saturated conditions will take over (e.g. sedges, pinrushes, dock).

Prevention:

- Management of drainage lines for efficient water flow.
- Management of surface water-flow to avoid surface ponding.

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- Plants species more tolerant of saturated conditions will take over (e.g. sedges, pinrushes, dock).

Prevention:

- Increase deep rooting vegetation for greater utilisation of water from the soil.

In India waterlogging is prevalent in the coastal areas of Mumbai and Kerala, estuarine deltas of Ganges, Andaman and Nicobar islands.

4.6.7 Sustainable Agriculture

In every agricultural region of the world, negative impacts of agriculture are being felt. The effects of soil degradation and agricultural pollution are more severe in some areas than in others, but there is universal recognition that while food production must increase to meet the needs of a growing population, the productive capacity of agricultural lands must be preserved and nurtured. Farm practices that seek to balance production and preservation are known as *sustainable agriculture.*

Although it is difficult to define precisely which farming techniques constitute sustainable agriculture, they generally involve intensive soil conservation measures and minimal use of pesticides and inorganic fertilizers. At the present time, the number of farmers using these methods is relatively small, mostly in specialized crops such as fruits and vegetables. But the number of farmers producing grain, meat, and dairy products using sustainable methods is increasing, and this trend is likely to continue as long as the costs of chemical inputs remain high and the need to maximize yield per acre is low.

One key feature of the drive toward sustainable agriculture is the reduced use of farm chemicals, especially pesticides. Pesticids have been a boon to modern technological agriculture, but they also have harmful side effects. The side effects of most concern are health hazards to agricultural workers using the pesticides, health effects on the general population through contamination of food, water, air and adverse ecological effects.

The most severe human health hazards of pesticides are those associated with the occupational exposure of farm workers handling the substances. Workers in the field at the time of application are exposed, as are those handling crops at harvest time. Accidental exposure is also a major concern. Although pesticides have been regulated to limit effects beyond the farm as well as on it, major problems remain. For example, nonpersistent pesticides that are generally used today do not accumulate in high concentrations in the environment. However, they are highly toxic at the time they are applied, hence those in contact with pesticides at that time are most at risk.

The general population is also exposed to agricultural chemicals, primarily through consumption of foods containing these substances but also through transport of pollutants in water and air. Although these exposures are less acute than those faced by agricultural workers, the number of people affected is much greater.

The combined hazards of pesticide use are great, but so too are the benefits in terms of increased yields. Concern for the problems associated with massive pesticide use has prompted research on alternative methods of pest control, most importantly integrated pest management. This approach recognizes that several different means can be used to control pests, including pesticides, crop rotation and other habitat controls, biological controls such as predator introduction, and other techniques. No single technique is likely to be completely successful in any given place, but for each particular set of agricultural needs and pest problems it should be possible to use a mix of different control techniques tailored to the situation. This approach will require considerable research and development before it can be widely used, but it offers the greatest promise in solving pest problems without poisoning humans or the environment.

4.7 ENERGY RESOURCES

Energy is an indispensable requirement in modern life. It is involved in all developmental activities of the world. Abundant and dependable sources are vital for the modern economic activities. The amount of energy consumption increases with increase in the per capita income of the country. All the amenities of modern world which are making our lives easy and comfortable are energy driven.

There are several sources of energy like coal, petroleum, natural gas, solar energy, wind energy and hydro energy etc. These sources are divided into two categories:

(i) **renewable** (an energy source that can be replenished in a short period of time)

(ii) **non-renewable** (an energy source that cannot be replenished in a short period of time).

Renewable and non-renewable energy sources can be used to produce secondary energy sources including electricity and hydrogen.

Sources of energy can also be categorised as: Conventional and non-conventional.

(i) **Conventional** (Coal, petroleum, natural gas and electricity (both thermal and hydel)

(ii) **Non-conventional** (solar, wind, tidal, geothermal, atomic energy and biogas)

On the basis of use, there are two categories:

(i) **non-commercial** (biofuels e.g. firewood, cow dung, charcoal and agricultural waste).

(ii) **Commercial** (coal, petroleum, natural gas, hydro-electricity and nuclear energy.)

4.7.1 Renewable Sources of Energy

4.7.1.1 Solar Energy

India, being a tropical country has enough scope for production and utilisation of solar energy. Our country is fortunate to receive solar energy for greater part of the year. This is an enormous energy resource, which is clean and pollution-free. It requires to be converted into other forms of energy by suitable technologies.

It has been estimated that India receives solar energy equivalent to more than 5000 trillion kWh (5000×10^{18} kWh) during a year. Under clear cloudless sky conditions the daily average of solar energy varies from 4 to 7 kWh/m².

There are two main ways of using solar energy to produce electricity

(i) Solar cells

(ii) Solar thermal technology

(i) Solar cells

Solar cells are photovoltaic cells that turn sunlight into electricity in an environmentally clean manner. Solar photo voltaic (SPV) systems have emerged as useful power sources for applications such as lighting, water pumping, telecommunications and power plants for meeting the requirements of villages, hospitals, lodges etc. They could be installed in remote areas in forests and deserts where installation of electric cables are cost-prohibitive.

Solar power, with government subsidy (Department of Non-conventional Energy Source, DNES, Government of India) is being used in remote rural areas in West Bengal in the forms of solar lanterns, solar streetlight and solar pump (for irrigation). Solar powered small pumps are being used in Delhi, Haryana and Himachal Pradesh. Electricity in Ladakh in the Himalayas is generated using solar panels, and this has brought about a tremendous change in people's lives. There are several companies in India that offer solar home-lighting systems. One 100 × 50 cm solar panel can easily power eight lamps.

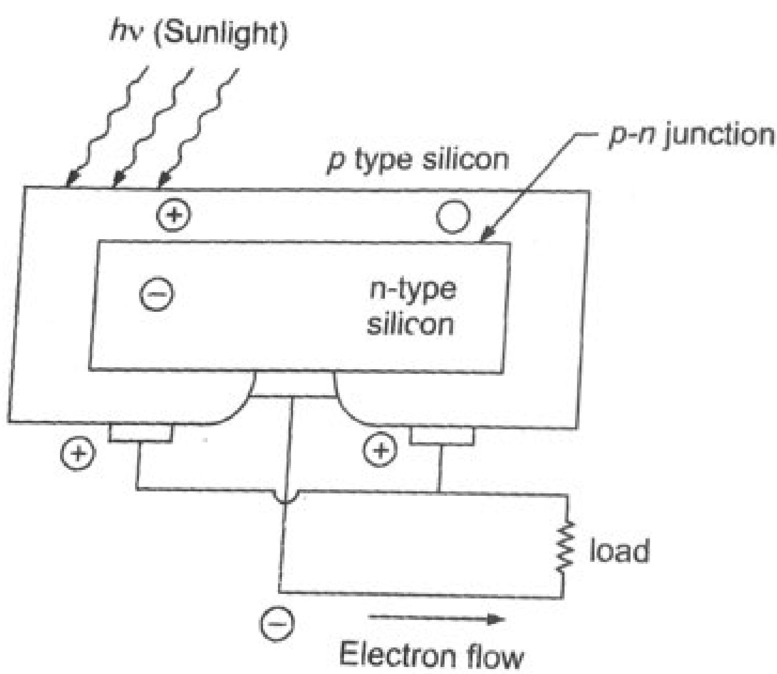

Fig. 4.6 Solar Cell for Electricity Generation.

Figure 4.6 illustrates the function of a solar cell. Light is absorbed in a plate, with the generation of positive and negative charges, which are collected at the electrodes on either side. The silicon solar cell, consists of a sandwich of a n-type and p-type silicon semiconductors (e.g. silicon, germanium is a crystalline substance which is intermediate between a metallic conductor on one hand and non-conducting insulator on the other). The charge separation is developed across the junction between them.

p-type silicon conducts positive charge while n-type silicon conducts negative charge. The silicon cell produces electricity but is quite expensive since very high-grade crystalline silicon is required for the cell.

The strength of current produced depends on the brightness of the light and the efficiency of solar cell. A potential difference (or voltage) of about 0.5 V is generated between the top and bottom surface of a solar cell. At present, the best designed solar cells can generate 240 W/m^2 in bright sunlight, with a maximum efficiency of about 25%.

The electric power generated by a single solar cell is quite small. Thus, to put solar cells to effective practical use, a large number of such cells are used together in the form of a solar panel. A solar cell panel consists of a large number of solar cells joined together in a definite pattern and provided with a protective encapsulation.

Through a solar cell panel (also called SPV system), solar energy is changed to dc electricity. This electricity can either be used as such or it can be stored in the batteries.

Merits of solar cells panels

- These are particularly useful for meeting the electricity requirements of decentralized applications. (As shown in figure 4.7)
- These are easy to install and maintain.
- These are noiseless, pollution free and have long life.

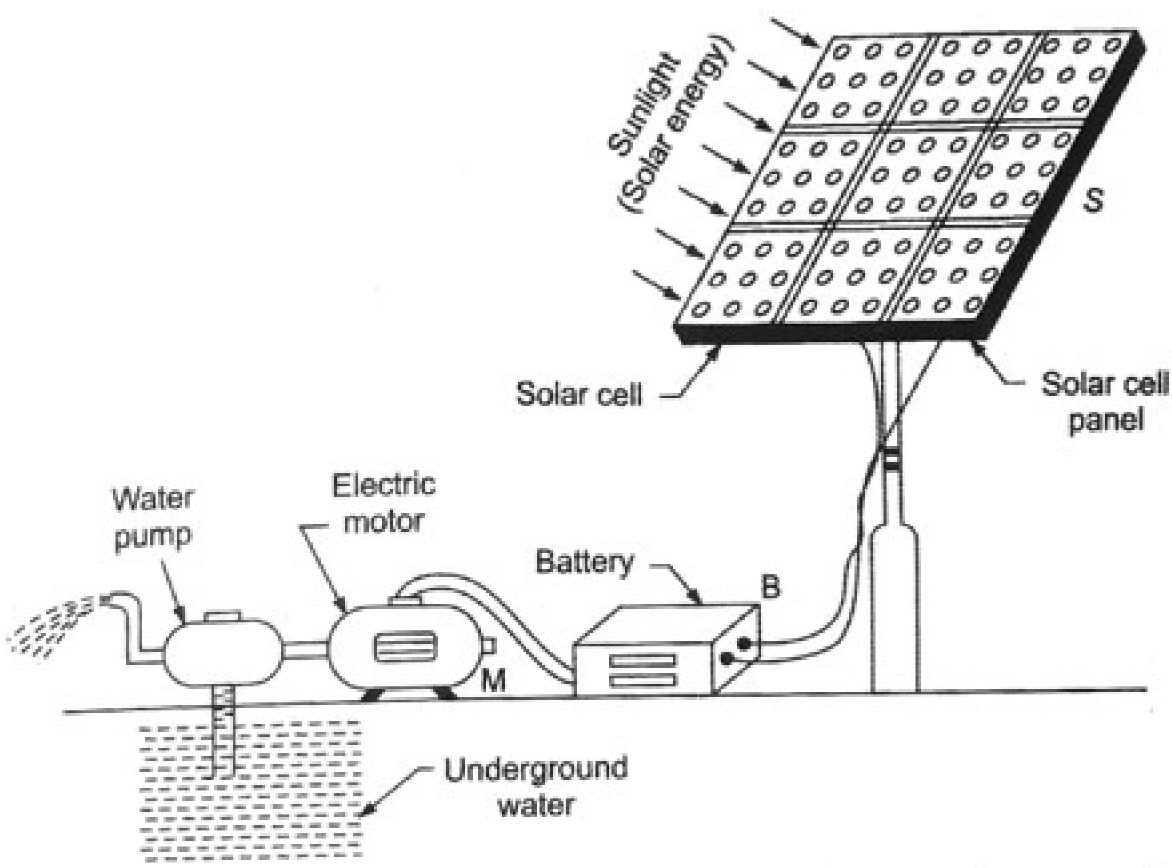

Fig 4.7 A solar cell panel is being used for running a water pump in a remote and inaccessible village where power transmission lines do not exist.

Limitations of solar cell panels

- The initial cost of installing an SPV system is high as silicon, wafer used in making solar cells and silver used in connecting solar cells to form a solar cell panel are costly.

- The electricity (dc) produced by solar cell panels is stored by charging dc batteries which are first to be converted to ac by using inverters.
- The efficiency of energy conservation is low as compared to other methods of generating electricity.

Developing countries such as Dominican Republic, Sri Lanka and Zimbabwe are leading users of solar cell panels.

(ii) Solar thermal technology

This uses heat gained directly from sunlight. The best known use of this technology is in:

1. Solar water heating
2. Solar heating of buildings
3. Solar dryer for food grains and other agricultural products
4. Solar distillation (for water purification)
5. Solar cooker (for cooking)
6. Solar powered vehicles
7. Solar thermal power generation

Solar thermal power generation

Of the various contributory factors to the global greenhouse effect, energy-related CO_2 emissions are the single largest source with power plants contributing about 11% of the CO_2 emissions. In this scenario, a cleaner energy technology like solar thermal power generation is imperative. Today, solar thermal power generation can competitively replace fossil fuel-based power generation at fuel price levels equivalent to US $100 per barrel, depending on radiation, infrastructure, and operational mode. Apart from grid-connected centralized power plants, solar thermal plants can also operate in stand alone mode in decentralized applications like rural electrification and in remote locations like islands. The world market is predicted to be 5.1 GW per year during 2010-15 including 800 MW per year potential in India.

4.7.1.2 *Wind Energy*

Atmospheric air is almost in a state of fast continuous motion due to unequal heating of landmasses and waterbodies by solar radiation. This kinetic energy possessed by air due to its velocity is called wind energy and can be used to do work. This energy was harnessed by windmills in the past to do mechanical work. For example in a water-lifting pump, the rotatory motion of windmill is utilised to lift water from a well. Today, wind energy is also used to generate electricity. A windmill essentially consists of a structure similar to a large electric fan that is erected at some height on a rigid support. To generate electricity, the rotatory motion of the windmill is used to turn the turbine of the electric generator. The output of a single windmill is quite small and cannot be used for commercial purposes. Therefore a number of windmills are erected over a large area, which is known as wind energy farm. The energy output of each windmill in a farm is coupled together to get electricity on a commercial scale.

Wind power potential of India

India is ranked fourth in the world for wind resource availability. The total wind power potential in India is estimated at 45,000 MW, out of which about 6000 MW is located in Tamil Nadu and 5000 MW in Gujarat. A total capacity of more than 1700 MW has already been installed which will increase further with the coming up of new facilities.

India's largest wind energy farm established near Kanyakumari in Tamil Nadu can generate 380 MW of electricity.

Various initiatives have been taken by the Ministry of Non-conventional Energy sources for the development of the wind power programme. The introduction of fiscal benefits and promotional incentives in terms of subsidy have attracted private investors which has provided a fillup for promotion of the technology. Guidelines for wheeling, banking and purchasing power from wind power projects have been sent to the states. It has been proposed that they may consider purchasing power at least at the rate of Rs. 2.25 per kWh. In addition to this, private sector can also avail loans from the Indian Renewable Energy Development Agency Limited (IREDA).

Merits of wind energy

- It is a non-polluting, environmental friendly and sustainable source of energy.
- The gestation period is low and the power generation starts immediately after commissioning of the windmill.
- Power generation is cheaper as there is no shortage of input (i.e., wind). The recurring expenses are almost nil.

Limitations of wind energy

- Wind energy farms can be located only in vast open areas in favourable wind conditions as the minimum velocity for a windmill to function is 15 km/h.
- The cost of construction of a wind energy farm is high.
- The appearance of windmills on the landscape and their continuous whirling and whistling is irritating.
- The location of wind energy farms should not be on the routes of migratory birds otherwise it will play havoc with the birds.
- There should be some backup facilities (like storage cells) to take care of the energy needs during a period when there is no wind.
- The tower and blades are exposed to rain, sun, storm and cyclone they need a high level of maintenance.

Note: IREDA was established in March 1987 as a public sector enterprise for the promotion, development and financing of new and renewable sources of energy technologies.

4.7.1.3 Hydroenergy

This renewable source of energy is produced from the kinetic energy of flowing water or the potential energy of water at height. This energy has been traditionally used for rotating the water-wheels and drive water-mills to grind wheat to make flour. This energy has been modified to generate electricity by establishing hydro-power plants. At hydro-power plants, the energy of falling water (or flowing water) is tapped by using a water turbine which drives the generator (Fig. 4.8). At present, the total electric power generated in our country, almost one-fourth is contributed by hydroelectricity. In India, if water resources are properly utilised, it may be possible to generate more than 10,000 megawatts of electricity. But at present only 6500 megawatts of hydroelectricity is generated. About 80% of the developed hydel projects lie in Maharashtra, Tamil Nadu, Karnataka, Kerala, Punjab, Himachal Pradesh, Jammu and Kashmir and Uttaranchal.

Advantages of hydel power Plants

- Hydroelectric power is pollution free and the most versatile source of energy out of all the known sources of energy.
- Hydel projects have a very low generation and maintenance cost, reliable and have a relatively long life. The life expectancy of a hydel power plant equipment is about 50 years or more.
- They have a very high efficiency over a considerable load and have quick start up and stopping time and rapid response to change in power demand.
- Hydel power enables us to conserve our coal resources as for every horse power (hp) of hydel power generated, about four metric tonnes of coal is saved.
- These projects are labour-intensive in nature, i.e., they enable us to tackle problem of unemployment.
- Hydroelectric power projects are multi-purpose projects as these enable us to use their water for irrigation, industrial and domestic purposes, control flood, develop recreational sites, etc.

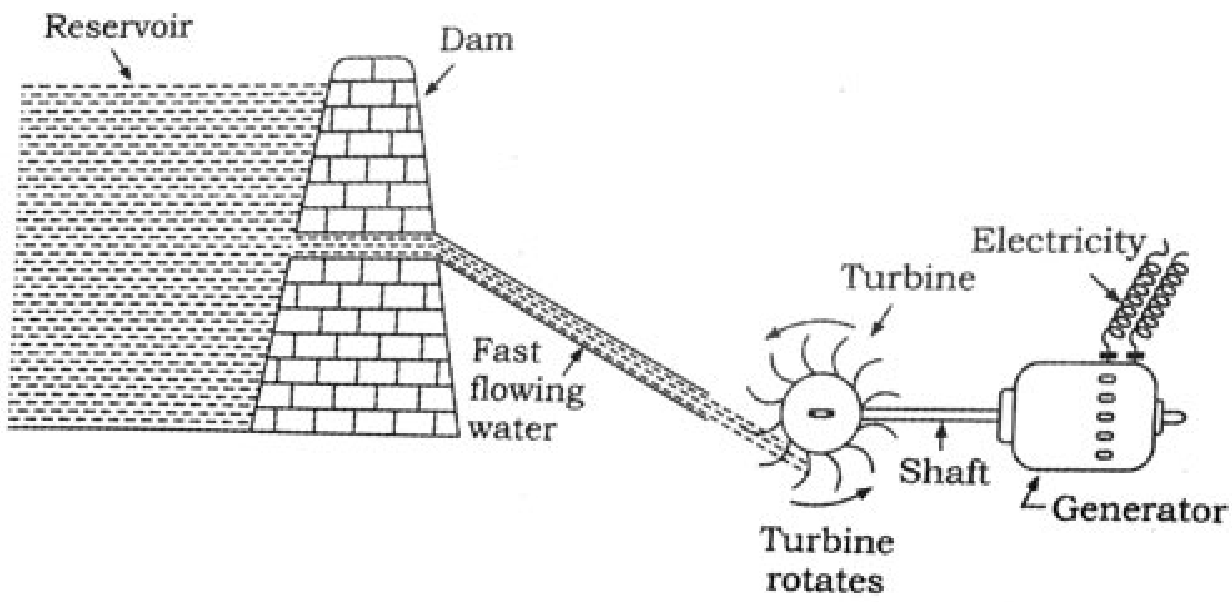

Fig. 4.8 Working of a hydro-power plant.

Demerits of Hydel Power Plants

1. The initial investment is very high and the gestation period is very long.
2. These projects cause population displacement, damage the environment and fertile land thereby creating a host of social problems as in the case of the Tehri Dam (on river Ganga) and Sardar Sarovar Projects (on river Narmada).
3. Hydroelectric generation is not suitable for all rivers and for all areas.

4.7.1.4 Biomass Energy

Biomass is the organic matter which is used as a fuel to produce energy. It is the oldest renewable source of heat energy still widely used as a fuel for domestic purposes. It includes fuelwood, agricultural wastes like crop residues, bagasse etc. (Bagasse is solid residual mass left after extracting juice from sugar cane) and cowdung. In India more than 70% of the rural population depends upon biomass as a source of energy. These are also burnt in rural industries and in the urban service establishments. It is common to use bagasse in the sugar industry and rice husk in rice mills. India is experiencing rapid economic and population growth leading to more energy requirement. The use of fuelwood alone in India is predicted to increase by a factor of three upto 2015.

These fuels however do not produce much heat on burning and a lot of pollutants like particulate matter, SO_2, NO_X and another trace gases are given out when these are burnt. Therefore, technological inputs to improve the efficiency of these fuels are necessary. Wood undergoes carbonization to produce charcoal which burns without flames, is comparatively smokeless and has higher heat generation efficiency.

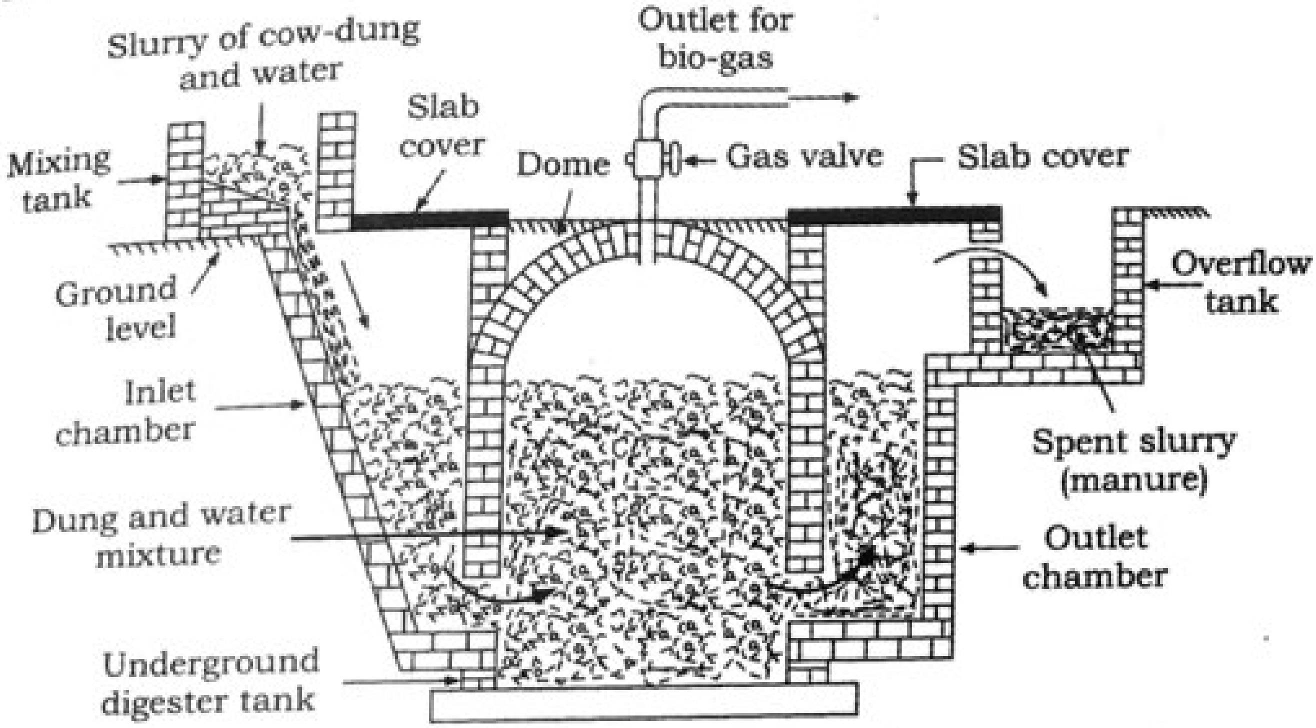

Fig. 4.9: Working of a Biogas plant.

Note: Carbonization is the process of heating the fuel in absence of air, to a sufficiently high temperature, so that the wood undergoes decomposition and yields a residue which is richer in carbon content than the original fuel. During this process moisture content and volatile matter is removed.

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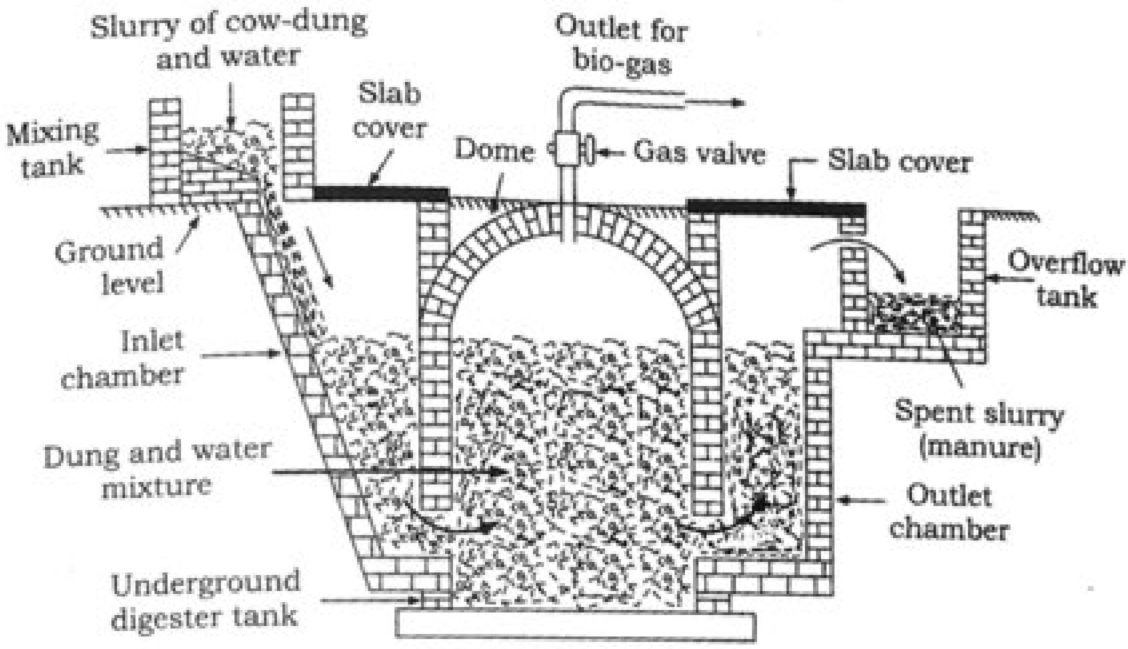

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Similarly cowdung, crop, residues, vegetable waste, poultry droppings and sewage are decomposed in the absence of oxygen (anaerobic decomposition) to give bio-gas (Fig. 4.9). It is an excellent fuel as it contains upto 75% methane. It burns without smoke, leaves no residue and has high heating capacity. The slurry left behind is removed periodically and used as excellent manure, rich in nitrogen, phosphorus and potassium. Biogas has a high calorific value (5000 - 5500 kcal/kg) and can be used for domestic purposes, operating small scale industries, lighting and as fuel for the boilers, generators etc. The large scale utilisation of biomass wastes and sewage materials for producing biogas provides safe and efficient method of waste-disposal besides supplying energy and manure.

Government's Initiatives Towards the Promotion of Biomass Energy:

(i) Biogas plants

The national project on biogas development was started in 1981/82 for the promotion of family-type biogas plants. The project aims at providing a clean and inexpensive energy source, producing enriched manure, improving sanitation, and elevating the status of women in rural areas. Against a potential of setting up 12 million plants based on cattle dung, about 2.7 million plants have been set up. A 10 MW rice straw based thermal plant has been commissioned by BHEL (Bharat Heavy Electricals Limited) at Jhalkhari in Punjab. A pilot plant to generate electricity from garbage and municipal wastes has been installed at Timarpur in Delhi.

(ii) Biomass gasifiers for thermal applications

The techno-economic feasibility of utilizing biomass gasifiers for a variety of thermal applications has also been established. The capacity rating of these gasifiers is up to 150 kg of biomass/hour (equivalent to 400000 kcal/hour energy). The lower capacity gasifiers are used in small-scale industries like silk reeling, dyeing, and drying and in community cooking, etc. The bigger sized gasifiers can be used in industrial applications such as carbon dioxide manufacturing, and magnesium chloride production, etc. Moreover, there is a demand for large gasifiers in industries like cement, sugar, and calcium chloride for meeting the process of heat requirement. The use of gasifiers offers substantial reduction in fuel consumption and reduction in harmful emissions. A 100 kW gasifier system has been established in Port Blair and a 15 kW sugar cane-water based system is under field evaluation.

(iii) Improved cookstoves

The National Programme on Improved cookstoves was launched in 1984/85 with the objectives of fuel conservation, reduction of smoke, conservation of forests and the environment, providing employment, and elevating the status of women and children in rural areas. Improved cookstoves have a thermal efficiency of 20% - 35% as compared to the traditional cookstoves which have an efficiency of 10%. In addition, the improved cookstoves (ICs) are estimated to save about 11 million tonnes of fuelwood equivalent every year. A minimum thermal efficiency of 20% and 25% has been prescribed for the fixed and portable type cookstoves, respectively. The pattern of subsidies for the promotion of ICs has been further rationalized to encourage market support.

4.7.1.5 *Geothermal Energy*

Geothermal energy is the heat of the earth and is the naturally occurring thermal energy found within rock formations and the fluids held within those formations.

In the molten core lying deep inside the Earth, the temperatures are as high as 4000°C. The thermal energy from this molten core forms an inexhaustible source of energy. Though the total quantity of heat stored in the Earth is vast, geothermal energy can be exploited only in particular areas, called the hot spots (e.g., volcanoes, geysers and bubbling mud holes). These hot spots are formed when geological changes push the molten rocks, called magma, upwards where it gets settled at some depths below the Earth's surface.

The underground hot water in contact with hot spots changes into steam. As the steam is trapped between the rocks, it gets compressed to high pressure. At some places, hot water and steam gush out from the Earth's surface after making their way through large cracks between the rocks and form natural geysers. Geothermal energy carried by natural geysers is utilized for generating electricity.

At some places, the steam trapped between the rocks is extracted by sinking pipes through holes drilled upto hot spots. The steam which comes out at high pressure is utilized to turn the turbine of an electric generator to produce electricity.

Although the most potent sources of geothermal energy are volcanos, hot springs and geysers, there are some other areas where geothermal energy can be harnessed under controlled conditions. This is done by pumping water down through an injection well where it passes through joints and fractures in hot rocks and then rises to the surface through a recovery or production well. This hot water is converted into steam by a heat exchanger. Dry steam is then passed through turbines to produce electricity.

Merits of geothermal energy

- Geothermal energy is the most versatile and least polluting renewable source of energy.
- It can be harnessed for 24 hours throughout the year and is relatively inexpensive.
- As compared to solar energy and wind energy, the power generation level of geothermal energy is higher.
- Geothermal energy can be used for power generation as well as direct heating.

Limitations of geothermal energy

- Geothermal hot spots are scattered and usually some distance away from the areas that need energy.
- The overall power production has a lower efficiency (about 15%) as compared to that of fossil fuels (35% to 40%).
- Though as a whole, geothermal energy is inexhaustible, a single bore has a limited life span of about 10 years.
- Noise pollution is caused by drilling operations at geothermal sites.

Geothermal power potential of India

India has vast potential for geothermal power. North-Western Himalayas and the Western Coasts are considered geothermal areas. The Geological Survey of India has already identified more than 350 hot water springs with average temperatures of 80°C–100°C. A 5 kW geothermal pilot power plant has been commissioned at Manikaran in Himachal Pradesh. The Puga Valley in Ladakh region has the most promising geothermal field with a potential of 4.5 MW power. A project on an experimental basis for the cultivation of mushrooms and poultry farming has been implemented.

4.7.1.6 Ocean Energy Systems

Oceans are large water bodies covering about 75% of the Earth's total surface area. Apart from being large reservoirs of water, they are huge reservoirs of energy also. Oceans store energy in many forms which can be obtained for useful purposes in the following ways:

(*i*) Tidal Energy
(*ii*) Wave Energy
(*iii*) Ocean Thermal Energy

(i) Tidal energy

Water level near the coasts rises up and falls twice a day. This movement of water level along the coasts is known as tides. Tides are due to gravitational pull of the Moon on waters in the ocean. This pull varies during the monthly cycle rotation of the Moon around the earth. High tides occur on every new moon day and full moon day and can raise the level of water by few meters. Tidal energy is harnessed by constructing a dam across a narrow opening to the sea. A turbine fixed at the opening of the dam converts tidal energy to electricity (Fig. 4.10)

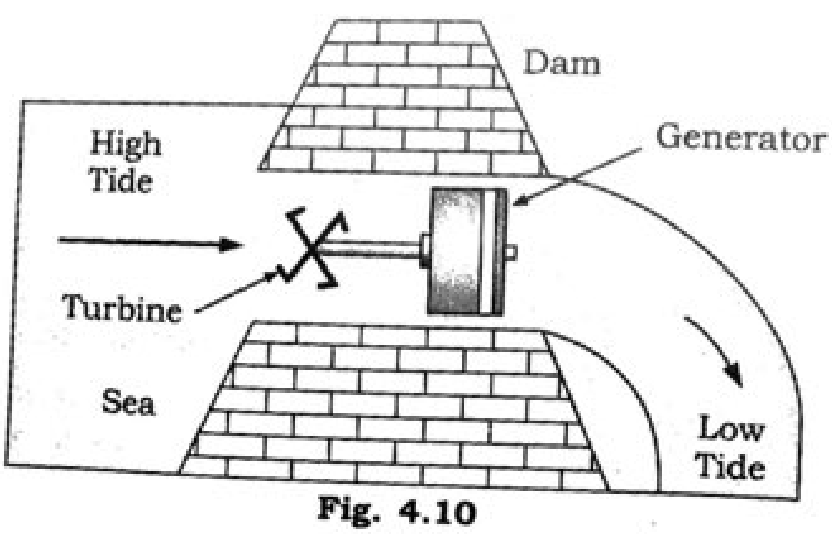

Fig. 4.10

Merits of tidal energy

- It is an inexhaustible, pollution free and renewable source of energy.
- It is independent of uncertainty of rainfall. Even if there is a continuous dry spell for many years, there is no effect on tidal power generation.
- A tidal power plant does not require large area of valuable land as it is built on the bay or the estuary.

Limitations of tidal power

- Due to variation in tidal range, the power output is variable and power generation is intermittent and not very large.
- There are very few suitable sites available for construction of dams.
- The most difficult problems in the use of tidal power is the barrage construction in areas of high tidal flow; and corrosion of barrage, sluiceways and turbines by salty sea waters.

Tidal power potential of India

India has an estimated tidal power potential of about 15,000 MW and the sites that have been identified are:

Gulf of Cambay (7000 MW), Gulf of Kutch (1000 MW) and Sunderbans (100 MW). Other suitable sites are near Lakshadweep Islands, Andaman and Nicobar Islands and Coasts of Orissa, Tamil Nadu, Karnataka, Kerala and Maharashtra. Asia's first tidal power plant costing Rs. 5000 crore and of 800 - 1000 MW capacity is proposed to be set up in the Hanthal Creek in the Gulf of Kutch in Gujarat.

(ii) Wave Energy

Due to blowing of wind on the surface of sea very fast water waves move on its surface. Due to their high speed, sea-waves have a lot of kinetic energy in them. The energy of moving sea waves can be used to generate electricity. A wide variety of devices have been developed to trap sea-wave energy to turn turbines and drive generators for the production of electricity:

(*a*) The floating generators are set-up in the sea. These move up and down with sea-waves and this movement drives the generators to produce electricity.

(*b*) The sea waves are made to move up and down inside large tubes. As the waves move up, the air in the tubes is compressed. This compressed air can be used to turn a turbine of a generator to produce electricity.

The total amount of power available in the World from wave energy is roughly 2 to 3 million MW. But the number of sites where this energy can be converted into useful energy is limited because only those areas with an average energy density of 40 MW/km of coastline are economically viable.

Merits of wave energy

- It is a free, renewable and pollution free source of energy.
- Wave power projects do not require large land areas.
- After extracting energy from the waves, the sea water is left in a relatively calm state.
- Wave power projects do not require a specific site (as in the case of tidal power projects) as same energy exists on almost any coastline.

Limitations of wave energy

- The power output is of variable nature.
- Wave energy extraction equipments must be capable of withstanding very severe peak stresses in storms.
- Wave power is expensive with the presently available technologies.
- Marine mammal and seabird population could be affected due to the presence of wave energy structures.

Wave power potential of India

The wave energy potential of 6000 km long Indian coast is estimated to be about 40,000 MW. Trade wind belts in Arabian Sea and Bay of Bengal are the ideal locations for harnessing wave energy. The first wave energy project with a capacity of 150 MW has been set up at Vizhinjam near Trivandrum.

(iii) Ocean Thermal Energy

Large amount of solar energy is stored in the oceans and seas. On an average, 60 million square kilometers of tropical seas absorb solar radiation which is equivalent to the heat content of 245 billion barrels of oil. If this energy can be tapped, a large amount of energy will be available to the tropical countries and the other countries as well. The sun warms the ocean water at the surface and the wave motion mixes the warmed up water downwards to the depth of about 100 m. This mixed warm layer is separated from the deep cold water layer and the temperature difference between these layers ranges from 10°C to 30°C. It is this temperature difference between the surface of the ocean and the depths of about 2 km which is used to produce electric power. The process of harnessing the ocean thermal energy is popularly known as ocean thermal energy conversion (OTEC). In the OTEC power plant, the warm surface water is used to boil a volatile liquid like ammonia. The vapours of the liquid are then used to run the turbine of generator. The cold water from the depth of the ocean is pumped up and condense vapour again to liquid.

Merits of ocean thermal energy conversion (OTEC) process

- The electric power from OTEC is continuous, renewable and pollution free. It is one of the most clean power production technologies.
- These systems transfer nutrients from the unproductive deep waters to the warmer surface thereby enriching the fishing grounds.
- OTEC process does not have daily or seasonal variations in their output as in the case with other solar energy devices.

Limitations of the process

1. This process requires a lot of capital investment.
2. Due to small temperature difference between the surface water and the deep water, the conversion efficiency is low (3–4%).

OTEC Potential of India

India has large potential of OTEC which could be of the order of 50,000 MW. Some of the best sites for this process plant are situated near the Lakshadweep, Andaman and Nicobar islands. An OTEC plant is proposed off the coasts of Tamil Nadu.

The energy potential from the sea (tidal, wave and ocean thermal energy) is quite large, but efficient commercial exploitation is still difficult.

4.7.2 Non-Renewable Sources

(a) Fossil fuels:

Fossil fuels are remains of prehistoric plants and animals which got buried deep inside the earth millions of years ago due to natural processes. These are energy rich compounds of carbon made by the plant and animal remains with the help of solar energy. Thus we can say that fossil fuels have stored solar energy in them.

(i) Coal

It is a complex mixture of compounds of carbon, hydrogen, oxygen and small amount of nitrogen and sulphur compounds. It is found in deep coal mines under the surface of earth. The coal found in the earth at different places has different percentages of carbon and so is classified according to the state of mineralization into the following types:

Types of coal	% Carbon content
Peat	11%
Lignite	38%
Bituminous	65%
Anthracite	96%

Lignite and bituminous coals burn faster and release a great deal of pollutants in the atmosphere on account of their lower carbon content. Anthracite burns slowly and releases much less smoke and delivers more energy and such is a best quality coal.

When coal is subjected to destructive distillation or carbonization (in the absence of air), then all volatile matter is removed from it to form coke. This residue is lustrous, dense, strong porous and coherent mass which is a better fuel than coal and does not produce smoke on burning. Also important and valuable by-products like tar, ammonia, naphthalene, benzol and H_2S are recovered during the carbonization process.

Coal mining in India dates back to 18th century. India has about 75118 mt which is approximate 7% of World's known coal reserves and these are mainly found in Tamil Nadu, Gujarat, Pondicherry, Rajasthan, Jammu and Kashmir, Bihar, Orissa, Madhya Pradesh. Coal supplies 50% of the country's total energy requirement. By current estimates, the reserves are sufficient to meet the country's demand for at least another 100 years. The largest coal consuming sectors are power, iron and steel and cement industries. The share of the thermal power sector in coal consumption is almost 75%.

(ii) Petroleum

Petroleum literally means rock oil. It is a complex mixture of several solid, liquid and gaseous hydrocarbons mixed with water, salt and earth particles. Small amounts of other compounds of carbon containing oxygen, nitrogen and sulphur are also present in petroleum. It is a thick black liquid and is not used as a fuel in its natural form. It undergoes fractional distillation to obtain a number of useful products like fuel oil, Diesel oil, Kerosene, Petrol or Gasoline, Petroleum Gas, Asphalt, lubricating oil and paraffin wax. The aggregate consumption of petroleum products upto year 2000 was 85 mt. The largest consumers of petroleum products are the transport, residential and industrial sectors.

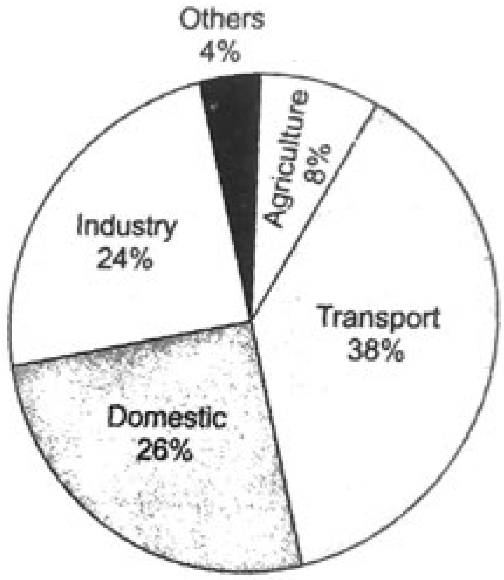

Fig. 4.11: Sectoral consumption pattern of petroleum products.

Source: Tata Energy Research Institute Energy Data (TEDDY) directory and Year book 2012–13.

(iii) Natural Gas

It is usually found underground near an oil source. However, there are some oil wells which give only natural gas. It is a mixture of methane (about 95%), ethane, propane and butane. Other components found are carbon dioxide, helium, hydrogen sulphide and nitrogen. It is highly inflammable, odourless, colourless gas. It is the cleanest burning fossil fuel found around the world but its largest reservoirs are in the former Soviet Union and the Middle East. Compressed Natural Gas (CNG) in liquid form is used as a fuel in transportation. It is a good alternative to petrol and diesel in reducing air pollution. It has a high calorific value of about 55 kJ/g.

Natural gas currently accounts for about 8% of the energy consumption in our country. The current demand is 89 mcmd as against domestic availability of 63 mcmd. The total gas consumption is around 20 bcm with a sectoral distribution as illustrated in Fig. 4.12.

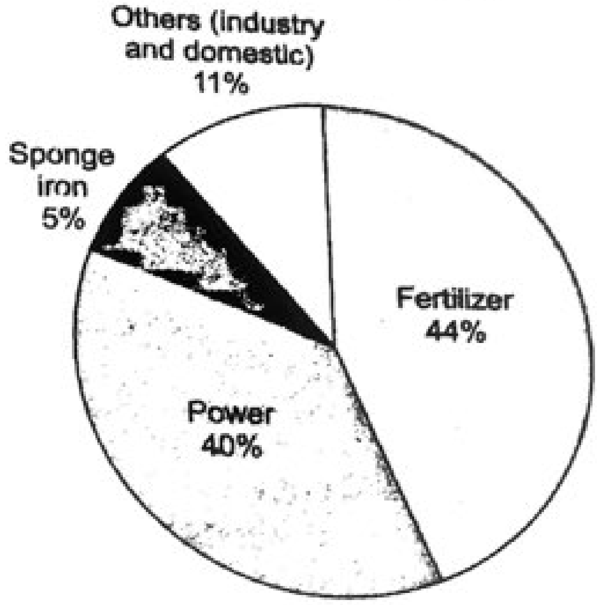

Fig 4.12: Natural gas end-use pattern (sectorwise).

Source: TEDDY, 2012–13.

The aggregate natural gas production is about 25 bcm and is likely to peak in future. However a wide gap is expected to prevail unless major gas discoveries materialize following future exploration by PSU's and private companies.

Also Liquefied Natural Gas (LNG) imports can ease the shortage of primary energy supply provided it can be sourced at competitive prices with respect to domestic and imported alternative fuels. In India, LNG is being imported from Qatar.

4.7.3 Nuclear Energy

Nuclear energy is released when atoms undergo nuclear fission. When heavy atoms like uranium, plutonium or thorium are bombared with low-energy neutrons, a tremendous amount of energy is released. The fission of an atom of uranium for example, produces 10 million times the energy produced by the combustion of an atom of carbon from coal. Currently all commercial nuclear reactors are based on nuclear fission. But there is a possibility of nuclear energy generation by a safer process of nuclear fusion. Sun derives its energy from the fusion of hydrogen nuclei into helium nuclei, which is going on inside it all the time. In a nuclear reactor designed for electric power generation such nuclear fuel could be a part of a sel sustaining fission chain reaction that releases energy at a controlled rate The released energy can be used to produce steam and further generat electricity.

Nuclear power reactors located at Tarapur (Maharashtra), Rana Prata Sagar (Rajasthan), Kalpakkam (Tamil Nadu), Narora (UP), Kaprapar (Gujara and Kaiga (Karnataka) have the installed capacity of less than 3% of th total electricity generation capacity of our country. However many industrialised countries are meeting more than 30% of their electric power needs from nuclear reactors.

Advantages of Nuclear Energy

- It produces a large amount of useful energy from a very small amount of nuclear fuel (like uranium-235).
- Once the nuclear fuel is loaded into the reactor, the nuclear power plant can go on producing electricity for two to three years at a stretch. There is no need for putting in nuclear fuel again and again.
- It does not produce gases like carbon dioxide which contributes to greenhouse effect or sulphur dioxide which causes acid rain.

Disadvantages of Nuclear Energy

- The waste products of nuclear reactions (produced at nuclear power plants) are radioactive which keep on emitting harmful nuclear radiations for thousands of years. So, it is very difficult to store or dispose off nuclear wastes safely. Improper nuclear waste storage or disposal can pollute the environment.
- There is the risk of accidents in nuclear reactors (especially the old nuclear reactors). Such accidents lead to the leakage of radioactive materials which can cause serious damage to the plants, animals (including human beings) and the environment.

4.7.4 Alternate Energy Sources - Green Fuel

The recent excitement over the Kyoto protocol pertaining to global climate change, the rising prices of fossil fuels has led India and other developing countries to look for an environmentally friendly and economically viable alternative fuel. Several initiatives have been taken in recent times on the energy front such as large scale promotion of wind energy farms for power generation, blending of ethyl alcohol with petrol and developmental efforts towards bio fuels or green fuels.

Biofuels include:

1. Vegetable oils (rape, hemp, sunflower, soy, palm, olive etc.)
2. Biodiesel (obtained from oil extracted from Jatrapha Curcas seeds.) Production of Biodiesel is two step approach; the extraction of the Jatropha oils from the seed, and the conversion of the extracted oil to Biodiesel, according to the following transesterification reaction.

   Oil + 3 Methanol [using NaOH or KOH catalyst = 3 Biodiesel (Methyl Ester) + Glycerol]

3. Ethanol (obtained by distillation from energy crops like sugarcane, maize, switch grass and populous tree etc and residues)
4. Methanol (wood alcohol obtained by pyrolysis from energy crops or waste)
5. Methane (gas obtained by bacteriological decomposition from landfill sites, sewage sludge etc.)

Carriage, storage and handling safety

Vegetable oils and biodiesel are ***biodegradable, non-toxic, non-carcinogenic, non-mutagenic and non-allergenic***. This is of considerable ecological advantage when considering the pollution that has been caused by petrofuels following large-scale marine oil spills - vegetable oils having been used to clean up oil slicks. Flash point* is around 179°C (diesel fuel 74°C), which means that the ***transport, storage and use is easier*** as the fire hazard is far below any other currently available liquid transport fuel.

Both ethyl and methyl alcohols are of similar safety hazard. Highly volatile (BP 78.3°C; 64.6°C; petroleum 27°C), flashpoints are also low, but comparable with petroleum for fire and explosion hazard.

Methane is a naturally occurring greenhouse gas, which is also the major component of CNG. It requires only 0.29 mJ energy to ignite, with an explosive concentration in air of 5 to 13%. This is 15 times the energy required to ignite a hydrogen/air mixture, but far lower levels of concentration are required. It is non-toxic and carries the same safety precautions as any of the liquefied gases currently dispensed from pressurised containers.

Environmental Effects

Growing biofuels should be considered essentially non-polluting in the longer term, given proper agricultural and process management. In addition by absorbing carbon dioxide from the atmosphere, ***energy crops act as a carbon sink*** throughout the year. This is released only when the fuel is used or the by-products burned. ***Longer term carbon sequestration*** (as discussed in Unit III) ***is achieved*** through the roots and stubble of the plants being ploughed into the ground.

There is a wide range of plants available, world-wide, as oil-producing energy crops. Other major benefit of energy crops is the oxygen by-product from photosynthesis.

Ethanol, methanol, vegetable oil and biodiesel are all oxygenated fuels. That is, they contain oxygen in their molecular structure. This is significantly advantageous to the burning process in an internal combustion engine, not only in improving the efficiency of the fuel but also in ***turning noxious oxides into less harmful dioxides*** (carbon and nitrogen). This also applies when used as fuel extenders.

In addition, none of them contain any sulphur at all unless minute traces are acquired through crop dusting, acid rain during growth or contaminated alcohols used during manufacture. Biodiesel is a better burning fuel than petrodiesel, and ***has a higher cetane number***. This is due to the molecular oxygen content of the fuel, which reacts under pressure and temperature faster than the subsequent fuel/air mixture. This causes the fuel to burn at a higher temperature, resulting in the formation of a higher level of nitrogen oxides. By simply retarding the fuel injection timing by two to three degree (depending on the engine design), the burning temperature is reduced, thereby reducing the amount of NOx gases formed.

***Flash point:** The lowest temperature at which the vapour of a flammable liquid ignites in air. Flash point is generally lower than the autoignition temperature.

Energy Production

Vegetable oil is the most energy positive fuel to produce requiring only agricultural input and seed pressing. 1 tonne of oil produced from rape seed per hectare gives an energy yield of 11 MWh at a cost of 2 MWh, a positive transport energy yield of 550%.

Biodiesel requires an energy input of up to 1 MWh per hectare, a positive transport energy yield of over 375%.

Since the importance of green fuels as alternate energy sources has been well recognised in India, the various organizations such as Indian Oil Corporation and Indian Railways, as well as several Universities and private institutions have initiated technology development plans for the use of bio-fuels. Many states like Tamil Nadu, Rajasthan, Karnataka and Andhra-Pradesh governments have introduced schemes to encourage investments in Jatropha plantations. The government plans to introduce 10% ethanol blended petrol from October 2008 instead of 5% blend that is currently mandatory. But Tata Motors have indicated to Ministry of shipping, road transport and highways that the higher blend would impact vehicles designed before 2002 and reduce fuel efficiency (*Hindustan Times*, Dec. 15, 2007). *The technical barrier to the encouragement of bio-fuels as a transport fuel is lack of suitable engines.* Despite the fact that Dr. Rudolph Diesel ran his new engine over a hundred years ago on peanut oil, subsequent engineering developments have been based on petroleum fuel. Thus India has to work and find out path to an ecologically sound, energy efficient future.

4.8 LAND RESOURCES

Land is the most important natural resource. It forms about 1/5th of earth's surface covering approximately 13, 393 million hectares. It supports natural vegetation, wildlife, human life, economic activities, transport and communication systems. The increasing human population has put a great pressure on the land resource, which is of finite magnitude. Therefore it is important to use the land resources with proper planning.

4.8.1 General Topography

India's mainland comprises four broad geographical areas: the Northern Mountains which has the great Himalayas, the vast Indo-Gangetic plains, the Southern (Deccan) Penninsula bounded by the Western and Eastern Ghats, and fourthly, the coastal plains and islands.

(*i*) **Northern Mountains:** Corresponding with the Himalayan Zone, alongwith country's northern boundaries including the Jammu and Kashmir (J&K), Himachal Pradesh (H.P.), North-West Uttar Pradesh (U.P.), Sikkim, part of Assam, and the North-Eastern States of Arunachal Pradesh, Nagaland, Manipur, Mizoram, Tripura and Meghalaya. The Himalayas comprise of mountain ranges which form an indomitable physical barrier as the world's biggest and largest mountain range. The Himalayas also contain the cold arid deserts and fertile valleys.

(ii) **The Great Plains:** Also known as the Indo-Gangetic plain is formed by the basin of three distinct river systems - the Indus, the Ganga and the Brahmaputra. The Plains extend from Rajasthan in the West to Brahmaputra valley in the East. The region covers the entire States of Punjab, Haryana, and the Union Territory of Chandigarh and Delhi and major parts of U.P., Bihar, West Bengal, and parts of Assam. These plains comprise one of the world's greatest stretches of flat and deep alluvium and are among the most densely populated areas of the world (456 persons per sq.km). The desert region, which contains the Great Thar desert, extends from the edge of Rann of Kutchh to larger parts of Rajasthan (Western) and lower regions of Punjab and Haryana.

(iii) **The Deccan Peninsula:** This zone covers the whole of South India which includes the States of Tamil Nadu, Karnataka, Andhra Pradesh and Kerala. The Region also covers the State of Madhya Pradesh, and parts of Bihar, Orissa, Puriliya district of West Bengal. Density of population is 202 persons per km^2. The Indo-Gangetic plains and the peninsular plateau are separated by mountain and hill ranges known as the Aravali, Vindhya, Satpura, Ajanta and Maikala ranges.

(iv) **The Coastal Plains and Islands:** The peninsula is flanked on either side by the Eastern Ghats and the Western Ghats. On either side of the Ghats outward to the sea lies a coastal strip. The Western coastal plain lie between the Western Ghats and the Arabian sea in the West, whereas the Eastern Coastal Plains face the Bay of Bengal in the East. This is also a region with very high-density population (349 persons per km^2).

The country's geographical area of 328 mha covers only 2.4% of the world's total area, on which 16.7% of the world's population and about 18% of the world's livestock population survive. Of the total area of 328 mha, landuse statistics are available for roughly 305 mha accounting for 93% of the total land area. Of this, roughly 264 mha of land is available for agriculture, forestry and related purposes.

4.8.2 Land Utilization

The land resources are utilised for the following purposes (Table 4.4)

1. Forests
2. Area under non-agricultural use e.g. land occupied by buildings, roads and railways, housing, recreational purposes, industrial sites and irrigation systems.
3. Barren and unculturable land, are generally unsuitable for agricultural use either because of the topography or because of their inaccessibility e.g. desert areas in Rajasthan, the saline lands in parts of Rann of Kutchh in Gujarat, the weed infected and ravine lands in Madhya Pradesh and alkaline lands in Uttar Pradesh.
4. Permanent Pasture and Grazing lands.
5. Fallow lands are left uncultivated from less than a year to five years or more.

6. Cropped areas: Area sown more than once in an agricultural ye plus the net sown area.

Table 4.4 Statistical distribution of total land area in India

A. Total Land Area	328.7 mha
Area which is enumerated in the census	306.50 mha
Forests	67.0
Area under non-agricultural use	21.8
Barren and Unculturable Land	19.4
Permanent Pasture and Grazing Lands	12.0
Fallow Lands	24.0
Cropped Area	142.5
— Area under Food Grain Cultivation	123.5
— Of this, Area under Rainfed Farming Systems	89

(Source: *Census of India, 2001)*

4.8.3 Land Degradation

The growth in the population and developmental activities have not only brought about degradation of land but have also aggravated the pace of natural forces to cause damage to land. At present there are about 130 million hectares of degraded land in India. Approximately 28% of it belongs to the category of Forest degraded area, 56% of it is water eroded area and the rest is affected by saline and alkaline deposits.

The terms land degradation and soil degradation are synonymous. Soil is integral part of land, hence, any deterioration in its quality, mass or volume, either singly or in combination, is also a deterioration of land. The term soil degradation is more specific and is directly related to crop production, while land degradation is a more comprehensive term.

Causes of Land Degradation

If natural hazards are left aside, the causes of land degradation can be divided into *direct* and *underlying* causes. Direct causes are inappropriate land use and unsuitable land management practices, e.g. the cultivation of steep slopes without soil conservation measures. Underlying causes are the reasons why these inappropriate practices take place, e.g. the slopes may be cultivated because the landless poor need food, and conservation measures are not taken because farmers lack security of tenure.

There are four direct causes of land degradation:

- deforestation and removal of natural vegetation;
- over-exploitation of wood cover for domestic use;
- overgrazing;
- agricultural activities.

Deforestation is a cause of degradation when the land that is cleared is steeply sloping or has shallow or easily erodible soils, and when clearance is not followed by good management.

Overcutting of vegetation occurs when people cut forests, woodlands and shrublands to obtain timber, fuelwood and other products at a pace exceeding the rate of natural regrowth. This is frequent in semi-arid environments, where fuelwood shortages are often severe.

Overgrazing is the grazing of natural pastures at stocking intensities above the livestock carrying capacity; the resulting decrease in the vegetation cover is a leading cause of wind and water erosion.

Agricultural activities that can cause land degradation include shifting cultivation without adequate fallow periods, absence of soil conservation measures, cultivation of fragile or marginal lands, unbalanced fertilizer use, and a host of possible problems arising from faulty planning or management of irrigation.

The role of population factors in land degradation processes are the underlying causes. In India, it is indeed one of the two major basic causes of degradation along with land shortage, and land shortage itself ultimately is a consequence of continued population growth in the face of the finiteness of land resources.

Population pressure also operates through other mechanisms. Improper agricultural practices, for instance, occur only under constraints such as the saturation of good lands under population pressure which leads settlers to cultivate too shallow or too steep soils, plough fallow land before it has recovered its fertility, or attempt to obtain multiple crops by irrigating unsuitable soils.

There are major soil problems in India which cause the degradation of soil. This soil degradation in general contribute to land degradation. These factors are:

(i) *Soil Erosion*

This is the process by which top soil is detached from land and either washed away by water, ice or sea waves or blown away by wind. This proces is also referred to as the creeping death of land. The processes of soil formation and erosion go on simultaneously and generally there is a balance between the two. This balance is disturbed due to the following human activities:

(a) **Deforestation:** Removal of vegetation cover has caused widespread erosion in Western Ghats, Uttar Pradesh and in Himachal Pradesh.

(b) **Faulty Cultivation methods:** For instance, in the Nilgris, land has been opened for cultivation of tuber crops like potato and ginger without undertaking anti-erosion measures like tracing of slopes. Also forests on slopes have been cleared at places to make way for plantation crops. Such faulty cultivation methods have caused soil erosion. Landslides are common feature in these areas.

(c) **Shifting Cultivation:** An ecologically destructive and uneconomic cultivation method is slash and burn or shifting cultivation which is practiced in hill areas of North-East, Chhotanagpur, Orissa, Madhya Pradesh and Andhra Pradesh. Vast areas have suffered erosion of soil in hill areas of North-East because of shifting cultivation.

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(d) **Overgazing:** A surplus of livestock population in our country is a big strain on grasses and fodder. The tread of cattle hardens the soil and prevents new shoots from emerging. Overgrazing by goats is a serious problem in certain stretches of the Aravalis and in Punjab and Himachal hills. The goats not only pull off leaves and branches, but they also uproot grass, as opposed to sheep, which only nibble the top shoots.

(e) **Diversion in Natural Drainage Channels by Railway Embankments and Roads:** Railway tracks and roads have to be constructed in such a manner that they are at a higher level than the surrounding area. But sometimes, road and rail embankments come in the way of natural drainage channels. This causes water logging on one side and water loss on the other side of embankments. All these factors contribute to erosion in one way or the other.

(f) **Lack of Proper Surface Drainage:** Because of lack of proper lack of drainage, water-logging occurs in low lying areas which lossens the top-soil and makes it prone to erosion.

(g) **Denuding Forest Fires:** These fires, sometimes natural but often man made, are very destructive. As a result of these, the forest cover is lost forever and soil is exposed to erosion.

(ii) Loss of Fertility by Mismanagement

The most important cause of soil degradation specially in tropics is the loss of fertility. In general, the loss of fertility by mismanagement includes - unscientific cropping practices, imbalancing of nutrients, nonjudicious use of fertilizers, loss of organic matter content and soil pollution.

(a) **Unscientific cropping practice:** Man, in his urge to derive the maximum yield to satisfy vast needs of a rapidly growing population, has been resorting to various scientific inputs like irrigation, fertilisers, pesticides etc. At the same time, unscientific cropping practices are sometimes followed. As discussed earlier at some stretches of Western Ghats, commercial tuber crops like potatoes and ginger are grown on slopes, after clearing the forests. These unscientific farming practices and an excessive use of inputs result in problems like soil erosion, loss of natural nutrients, water-logging and salinity and contamination of ground and surface water.

(b) **Imbalancing of Nutrients:** A sudden rise in the use of chemical fertilisers and pesticides has harmed the long term fertility of soil and caused water pollution.

(c) **Loss of Organic Matter:** Loss of organic matter is a form of soil degradation, not so much because of the direct effects of organic matter, but more importantly because of the many indirect effects. In addition to the buffering effect on nutrient ion concentrations, and the complexing of potentially toxic ions, the indirect benefits include the stabilizing effects on soil aggregates, and the support offered to the fungal and microbial soil population.

(d) **Soil Pollution:** Soil pollution originates from the development of industry, use of pesticides, chemical fertilisers, use of sewage, sludge, city composts etc which is the major factor of loss of fertility.

(iii) Salinity/Alkalinity

This problem occurs in areas of temporary water surplus and high temperatures. Due to over-irrigation or high rainfall, the moisture percolates down and dissolves the underground salts in it. During the dry period, this solution comes to the surface by capillary action. The water gets evaporated, leaving behind a crust of salts of sodium, magnesium and calcium which has a fluorescent appearance. This salt layer plays havoc with the fertility of top soil and renders vast stretches of useful land infertile. This problem is particularly serious in areas with assured irrigation in Punjab, Haryana, Uttar Pradesh, Western Maharashtra, Bihar and Northern Rajasthan (the Indira Gandhi Canal command area). An area of around 7 mha suffers from the problem of salinity/alkalinity.

(iv) Acidity

It is a chemical degradation of soil including toxicity of certain metallic ions for the normal crop growth. Acid soils are characterised by low pH values, low base saturation and high amounts of exchangeable hydrogen and aluminium resulting from intense leaching. These soils cover an area of about 49 million ha. In high rainfall zones of Himalayan belt, north-eastern region, Western Ghats, Eastern Ghats and coastal belt of Kerala. Nearly 26 million ha of acid soils have pH values below 5.5, whereas 23 million ha have pH ranging from 5.5 to 6.0. The acid sulphate soils of Kerala have more of polysulphides in the form of pyrites, free sulphuric acid and sulphates of Fe and Al. In extreme case pH of sulphate soils drops below 2.0 with concurrent breakdown of clay lattice and release of free iron and aluminium.

(v) Waterlogging

Waterlogging is defined as the state whereby soil becomes saturated with water within the depth of the root zone for a period that affects yield and quality of crops.

This happens when the water table gets saturated for various reasons- over-irrigation, seepage from canals, inadequate drainage etc.

National commission on Agriculture reported an area of about 6 million ha affected by waterlogging over 17 states in India. Out of this 3.4 million ha area suffering from surface flooding mostly in the states of West Bengal, Orissa, Andhra Pradesh, Punjab, Uttar Pradesh, Gujarat, Tamil Nadu, Kerala and M.P. about 2.6 million ha has a problem of waterlogging owing to high watertable. However, central groundwater board reported that during monsoon period, an area of 36.36 million ha is temporarily waterlogged. The land under waterlogged conditions can be used neither for agriculture nor for human settlements. In dry areas, waterlogging leads to salinity and alkanity.

(vi) Deterioration of Soil Structure

Deterioration of soil structure is the commonest form of physical degradation. It is most commonly found where heavy machinery is used to cultivate the soil. Problems occurs mostly in soils of intermediate texture and low organic matter content. It involves: loss of stability of aggregates in surface soils, leading to sealing compaction and crusting, and consequently poorer infiltration rates and greater run off and erosion. Lack of organic matter and high percentage of fine silt are the factors contributing to sealing. Crusting is the formation of more compact, hard and brittle layer on the soil surface.

(vii) Floods and Droughts

Both these hazards have the harmful effect of limiting the use of good soil. This has already been discussed earlier. One nagging effect of floods is that each year a new area is affected.

(viii) Desertification

The clay and humus are the most important components of soil for both nutrient - and water-holding capacity. As clay and humus are removed, nutrients are removed as well, because they are bound to those particles. The loss of water-holding capacity is even more serious, however. Regions that have sparse rainfall or long, dry seasons support grass, scrub trees, or crops only so far as soils have good water-holding and nutrient-holding capacity. As these soil properties are diminished by the erosion of top-soil, such areas become deserts, both ecologically and from the standpoint of production. The term desertification is used to denote this process. Desertification doesn't mean advancing deserts; rather, the term refers to the formation and expansion of degraded areas of soil and vegetation cover in arid, semiarid, and seasonally dry areas, caused by climatic variations and human activities. These lands are traditionally called drylands.

The problem of desertification is serious in areas adjoining the Thar desert in Rajasthan. The affected areas lie in the states of Punjab, Haryana, Dehi, Madhya Pradesh and the Aravalis in Rajasthan.

Even though variations in climate often play a role in the processes of desertification, it is human agency that is the greatest threat to the dryland ecosystems. The three major practices that expose soil to erosion and lead to desertification are:

- **Overcultivation**
- **Overgrazing**
- **Deforestation**

4.8.4 Lands Slides

Landslides are among the major hydro-geological hazards that affect large parts of India, especially the Himalays, the Northeastern hill ranges, the Western Ghats, the Nilgiris, the Eastern Ghats and the Vindhyas, in that order. In the Himalayas alone, one could find landslides of every fame, name and description - big and small, long and short, quick and creeping, ancient and new.

India has a sensational record of catastrophes due to landslides. The Darjeeling floods of 1968 destroyed vast areas of Sikkim and West Bengal by unleashing spate of landslides, causing considerable death and destruction. These landslides occurred over a three-day period with precipitation ranging from 500 to 1000 mm in an event of a 100-year return period. The 60 km mountain highway to Darjeeling got cut off at 92 places resulting into loss of lives and total disruption of the communication system. Yet another landslide tragedy of an unprecedented dimension was the great Alakanada Tragedy in July 1970 in Uttaranchal State that resulted from the massive floods in the river Alaknanda, upon breach off a landslide dam at its confluence with river Patal Ganga. A few years ago, the Malpa rock avalanche tragedy in August 1998, instantly killed 220 people and wiped out the entire village of Malpa on the right bank of river Kali in the Kumaun Himalaya of the state of Uttaranchal.

The economic losses due to landslides in India are enormous and recurring.

Man as the agent of change

Who is responsible for the alarm in the areas affected by landslides? Let us take the example, the Himalayas. Today these face the greatest danger of landslides? Himalaya's immature geology, meandering rivers, snow bodies, climatic variations, cloudbursts, flash floods etc., have always been there. For centuries, formidable snow avalanches did hurtle down the slopes in the higher Himalaya as they do now.

The real chaos on slopes came when the man entered the scene. Vast areas of Western Sikkim, Kumaun, Garhwal, Himachal Pradesh, Kashmir and several other hilly regions fell to his axe and were robbed of the protective vegetal cover to less than 30 per cent as against twice as much considered desirable. *Cutting of trees for fuel or fodder, overgrazing, increased domestic and industrial consumptions of timber were chiefly responsible for deforestation.* For example, close to the India-Bhutan border near Phuentsholing (Bhutan) and Jaigoan (India), spurt of human settlements has increased the number and frequency of Landslides, over the period of past few decades. Deforestation did add to the fragile geological environment and adverse climatic conditions to cause degradation of inter bedded quartzite and phyllites. The slopes without vegetation could not be expected to hold soil cover together and widespread erosion was the natural consequence.

Nature takes nearly 1000 years to produce a few centimetre of top soil but destabilizing forces of nature in the mountainous areas wipe away millions of cubic metre in just one second. The rate of erosion in the catchment area of the Himalayan rivers has increased five-fold in the geological time scale; the present rate being upwards of 1 mm per year.

As the pressure of population rapidly grew, more and *more of human settlements, roads, dams, tunnels, water reservoirs, towers and other public utilities were added.* The network of roads in the Himalayan region is today well over 50,000 km. Some of the roads exist even at altitudes as high as

5,000 metre surrounded by mountain ranges such as Kanchenjunga (8,586 m). Khardung La at 5,600 m is perhaps the highest motor road in the world.

Construction of roads requires imaginative planning and methodical construction but when engineers work against time, they may not even have the basic data on geological formation, topography, drainage pattern etc. and as a result, *some of the hill roads begin with landslides as witness to their inauguration functions.*

Likewise, a number of dams have been built in the Himalaya including the mighty Tehri Dam on river Bhagirthi. The Dam projects over the Ganga and its tributaries in the hills alone exceed two dozen. Such constructions include large scale deforestation, huge excavations, resettlement problems and consequent threats to life and property. A number of tunnels are also being made. Microwave, TV, Transmission Line and other towers are also spread over the hilly areas. Quarrying and mining, for example, in the Doon Valley, Jhiroli (Almora) and Chandhak (Pithoragarh) have inflicted heavy damages to slopes and to the associated environment.

Influxes of tourists into the hilly region have brought about tremendous pressure on land due to construction of new buildings and tourist complexes including aerial ropeways and other utilities. For water storage, underground water tanks are being built without any special instructions or precautions despite considerable bad experience of slope instability from similar tanks, as at Mussorrie in Uttaranachal and Shimla in Himachal. Some of the constructions are coming up on old landslide masses without adequate pretreatment and investments on hillside stability, compounding the problem. Our rich cultural heritage and monuments are also under severe threat due to landslides and slope instability.

The time has come when the problem due to land slides should be recognised and Government should ensure that protective measures are taken for any development activity to start in the hilly region.

Source: R.K. Bhandari, The Indian Lanslides, Scenario, Strategic issues and action points, Indian Disaster Management Congress, New Delhi, November 2006.

EXERCISES

Short Answer Questions

1. Define the term 'resource'.
2. Distinguish between renewable and non-renewable resources with examples.
3. Briefly explain the importance of forests.
4. How are forests important in keeping an ecological balance?
5. What are the negative impacts of timber extraction?
6. Discuss the benefits and problems of dams.
7. Write a short note on common approaches towards water conservation.
8. Write short notes on the following

1. Floods	2. Droughts
3. Conflicts over water in India	4. Saltwater intrusion
5. Land subsidence.	

9. Write a short note on mineral resources of India.
10. Write a short note on the classification of minerals
11. Write short notes on
    (a) Surface and subsurface mining
    (b) Processing the mineral
12. Write short notes on
    (a) Green revolution
    (b) Water logging
    (c) Salinity
13. List the environmental problems associated with
    (i) chemical fertilizers, and
    (ii) pesticides
14. Write short notes on
    (a) Wind energy
    (b) Biomass energy
    (c) Hydroenergy
    (d) Ocean thermal energy
    (e) Nuclear energy
15. Discuss in brief the advantages and disadvantages of non-conventional sources of energy.
16. Write notes on
    (a) man induced landslides
    (b) desertification
    (c) water logging
    (d) salinity in soil.
17. (a) Distinguish between Under nutrition and Malnutrition
    (b) What are renewable, non-conventional energy sources? Give examples.
    (c) Why is recycling usually more efficient than mining new raw materials?
18. (a) What are the causes of deforestation in India? Describe briefly its effects on the environment.
    (b) What are the different types of energies which can be derived from ocean? Explain briefly.
    (c) What are the major approaches to conserve water resources? How is water harvesting practiced in India?
19. (a) Write short notes on:
    (i) Solar Energy
    (ii) Salinity and Water logging
    (b) What are Green Fuels? Why are these considered non-polluting and clean fuels?

Long Answer Questions

1. What do you mean by natural resources? Enumerate the various natural resources.
2. How can you classify the forests? What is the importance of forests in our life?
3. Explain the major causes of deforestation? Briefly describe the effects of deforestation on environment.
4. What do you mean by deforestation? Explain its causes and ill-effects.
5. What are the factors which have contributed to the conservation of forests in India?
6. Discuss the impacts of dams on the forests and tribal people.
7. What are the alternative energy resources? Differentiate between renewable and non-renewable natural resources.
8. Describe in detail the consequences of overexploitation of water resources.
9. Explain the types of drought and their inter-relationship.
10. Identify and explain the core causes of water crisis in the world.
11. Discuss the distribution and uses of mineral wealth in India.

12. What are the different methods adopted for the extraction of mineral wealth.
13. Write a note on environmental impact of extracting mineral resources.
14. What are the different types of minerals. Describe the effects of mineral extraction on the environment.
15. Discuss in detail the major food problems. How is under nourishment different from malnourishment?
16. Write a detailed note on negative impact of modern agriculture on environment.
17. Discuss the various types of agricultural practices.
18. What do you mean by water logging and salinization? What are their causes and how do they influence the agricultural yields? Suggest measures to minimise them.
19. Write a brief note on sustainable agriculture.
20. Name and explain in brief the various ways in which, energy from oceans can be obtained.
21. What is Geothermal energy? Discuss its merits and demerits.
22. Describe various non-renewable energy sources in detail.
23. Discuss the importance of green fuel as an alternate source of energy.
24. Discuss the various cause of land degradation.
25. Explain causes of soil erosion and enumerate the methods of prevention of soil erosion.